Random Feature Galerkin Methods for Accuracy-Matched Incompressible Flow: Discretization, GPU Kernels, and End-to-End Performance Attribution

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Abstract

We study numerical adaptations and low-level implementations of a fixed random feature Galerkin solver for incompressible flow, separating the effects of the trial space, time schedule, operator cost, and runtime overhead. For two-dimensional flow in a finite annulus without imposed reflection symmetry, we specify boundary-compatible features, weighted rank truncation, and a weak discretization with an independent mean-flow block. We derive a variable-step SBDF2 bridge with boundary lifting and establish its local consistency. Common coordinates for the time-phase matrices of each angular mode yield a geometric anchor that minimizes a stated worst-case condition-number bound. The implementation analysis describes FP64 FFT packing and shared-memory fusion, warp-level probe reduction, native C radial kernels, and setup and compilation-artifact reuse. At $Re = 100$ and $T = 72$, all 78 GPU trajectories, comprising 39 configurations under two cache policies, meet the prescribed 0.5% velocity and vorticity tolerances. Piecewise schedules remove 11.11% and 9.36% of true nonlinear evaluations. FFT fusion reduces short-trajectory marching time by 9%–17% but increases the first full-solve cost by 23%–69%. Twenty CPU trajectories supplement feature and switch controls. Both fixed-midpoint frequency configurations qualify with analytic layers, while all 12 configurations without nonzero-mode layers fail; retained rank also decreases, so this is not an equal-rank enrichment ablation. Delaying the switch only modestly improves accuracy at greater nonlinear cost, and radial/modal profiles distinguish the distribution of velocity and vorticity errors. Reduced Legendre spaces also qualify, and the final isolated timing campaign is not accepted; an overall speed advantage over comparably optimized traditional CFD therefore remains unestablished. The contribution is a reproducible combination of discrete derivations, source-level cost explanations, and scoped positive and negative evidence. Local consistency, transformed conditioning, and numerical qualification are not interpreted as guarantees of nonlinear stability or universal performance.

Keywords Random features, incompressible Navier–Stokes equations, Galerkin methods, variable-step SBDF2, GPU kernels, performance attribution

1 Introduction

Neural features provide alternatives to polynomial trial spaces for partial differential equations. A flow solver, however, combines a representation, spatial projection, time integration, linear algebra, nonlinear transforms, and runtime management. A speedup observed after changing several of these components cannot be attributed automatically to random features. This study examines that attribution problem for a fixed random feature Galerkin discretization of two-dimensional unsteady incompressible flow, and asks how its mathematical structure translates into measurable computational savings.

Unsteady wakes make the comparison demanding. Velocity involves first derivatives of the streamfunction, whereas vorticity involves second derivatives; a small velocity error can therefore

coexist with a substantially larger vorticity error. Alternating shedding also accumulates phase error. Independently shifting each trajectory in time can make plots agree without recovering the original initial-boundary-value solution. Meanwhile, a small GPU operator may take only tens of microseconds, even when compilation, module loading, FFT planning, and context initialization consume a substantial fraction of the full solve. We consequently require common physical accuracy, original time coordinates, and explicit end-to-end cost boundaries.

The test problem is a finite annulus at $Re = 100$, integrated to $T = 72$ without imposed reflection symmetry. The radial approximations use fixed random trigonometric features, deterministic Legendre features, or fourth-order spatial finite differences. All retain the full real Fourier representation in angle and solve the same physical problem with second-order time integration. Legendre–Galerkin is itself a traditional CFD baseline and must be included in that category. The fixed-feature solver is distinct from a PINN that trains a representation of an entire trajectory by residual minimization.

Three questions organize the study. First, how much actual work can piecewise time steps and smaller retained spaces remove while preserving both velocity and vorticity accuracy? Second, which costs do common phase coordinates, FFT fusion, and parallel observation change, and can other implementation costs offset the algebraic savings? Third, does the available evidence establish an overall advantage over strong deterministic baselines under matched accuracy, initialization policies, and timing boundaries? These questions require discrete analysis, operator experiments, and complete isolated runs, respectively; one local speedup cannot answer all three.

Table 1: Classification of the work and its evidential limits. A numerical adaptation is distinct from a new basic algorithm; a local speedup is distinct from a full-solve advantage.

Component	Category	Available evidence	Limit
Lifting and one variable-step bridge	Numerical-method adaptation	Consistency derivation and trajectory qualification	No unconditional stability theorem for the full NS system
Common phase coordinates and anchor	Linear algebra analysis	Same-mode identities and a spectral bound	Bound minimization does not minimize measured runtime
Retained radial spaces and Legendre control	Discretization and validation	Five fixed seeds and reduced deterministic spaces	No universal random-feature advantage
Fused FFT, warp observations, C kernels	Low-level implementation	Source, local tests, short trajectories	Application kernels, not a third-party FFT library patch
Shared setup, caching, startup path	Runtime engineering	Stage records and cache equivalence	Final isolated ranking remains unestablished

The principal results are verifiable workload reductions and performance attribution. The two time schedules remove approximately 11.11% and 9.36% of true nonlinear evaluations. All 78 GPU trajectories, covering 39 configurations under two cache policies, satisfy the registered numerical criteria. The specialized fused FFT is effective on fixed states and during short trajectory advancement, yet increases the cost of the first complete solve. The final isolated timing campaign failed its acceptance criteria, and its remaining records are not used to rank methods. These limits position the paper as numerical computation and application-level high-performance computing research. The evidence does not establish a general convergence theory or a universal speed advantage over traditional CFD.

Additional factorial ablations examine whether randomness is necessary: both fixed-midpoint frequency configurations qualify when analytic layers are retained, while all 12 configurations

without nonzero-mode layers fail the field criterion and also lose retained rank. The conclusion is therefore restricted to the effectiveness of the present enriched spaces, without automatically attributing it to random sampling. Section 9 adds radial/modal error decomposition and delayed-switch controls; Section 13.1 identifies the algebraic structure lost on nonseparable geometries.

2 Related work and scope

2.1 Random features, projection solvers, and trained representations

The random feature method fixes nonlinear features and determines their combination coefficients for PDE approximation[1]. Local extreme learning machines combine fixed hidden layers with domain decomposition[2]; randomized neural network Petrov–Galerkin methods have been applied to elasticity and Navier–Stokes equations[3]. High-precision RFM, randomized preconditioning, and orthogonal feature learning address accuracy, conditioning, and representation quality[4–6]. Random features combined with Galerkin projection, preconditioning, or orthogonalization are consequently not standalone novelty claims of this work.

Time-dependent RFM can also use space-time features and partitions of unity, with global or blockwise least-squares solves[7]. The present implementation takes a different route: random features define only the radial trial space, Fourier modes represent angle, and a stepwise IMEX scheme advances time. It neither learns temporal features nor repeatedly minimizes a trajectory-wide residual. Error statements for block propagation in space-time least squares therefore require a separate derivation before being applied to this SBDF2 evolution, and performance results do not transfer automatically between the formulations.

PINNs use automatic differentiation to construct residual losses and train network parameters[8]. Causal training changes the treatment of temporal learning, while PirateNets use adaptive residual connections to improve deep PINN optimization[9, 10]. A comparison with such methods must account for training, restarts, sampling, inference, and derivative reconstruction according to the intended use. Published errors and times from different geometries, boundaries, horizons, or hardware cannot be inserted into an accuracy-matched table here. No complete trained-model comparison on this same problem and tolerance is available in the present evidence. These references delimit the scope rather than establish a speed ranking.

2.2 Time discretization and linear algebra

IMEX multistep methods and variable-step extensions are established approaches[11, 12]. The second-order coefficients used here belong to that framework. Our analysis concerns their compatibility with the discrete mass, boundary lifting, contracted flux histories, and a single step-size transition in this implementation. Likewise, whitening and diagonalizing a symmetric definite generalized eigenproblem use standard linear algebra[13]. The relevant questions are whether multiple time phases of one angular mode can share coordinates, what an anchor selection actually optimizes, and why fewer factorizations can still increase setup cost when the original device interface is retained.

2.3 GPU fusion and application performance

FFT dealiasing, Hermitian representations of real fields, and mixed-radix transforms are established techniques[14, 15]. cuFFTDx already provides device FFTs, fused convolution examples, and comparisons with cuFFT[16]; warp exchange and reduction use standard CUDA programming mechanisms[17]. The engineering work here implements and checks these mechanisms for a particular FP64 multifield flux layout, including channel-scale sensitivity, startup overhead,

and unsuccessful candidates. It does not introduce the general idea of FFT fusion as a new FFT algorithm.

High-level languages can organize high-performance flow solvers whose computational kernels execute in compiled libraries or native implementations[18]. We therefore discuss bottlenecks through data movement, synchronization, matrix layout, and lifecycle cost. Choosing C, CUDA, or another host language should follow the measured cost fraction and the work that can actually be removed. The same reasoning applies to an upstream contribution: the present changes are application-side kernels, with no modification of third-party library source. A reusable upstream sample needs independent correctness, an equivalent library-based comparison, and a reproducible build; a local application speedup alone does not constitute a library performance fix.

3 Flow model and the fixed approximation space

3.1 A driven finite-annulus problem

We consider the two-dimensional incompressible Navier–Stokes equations

$$\partial_t \boldsymbol{u} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u} = -\nabla p + \nu\Delta\boldsymbol{u}, \qquad \nabla\cdot\boldsymbol{u} = 0, \qquad \Omega = \{(r,\theta) : a < r < R\}. \tag{3.1}$$

The cylinder diameter and the final streamwise speed define the length and velocity scales. The calculations use $a = 0.5$, $R = 6$, $\nu = 0.01$, and hence $\mathrm{Re} = 100$. The final time is $T = 72$. Introduce the logarithmic radial coordinate $s = \log(r/a)$, its extent $L = \log(R/a)$, and $z = s/L$. The area measure becomes $r^2\,ds\,d\theta$. The physical radial span is $R - a = 5.5$; it is not the logarithmic extent $L = \log 12$ used in the formulas.

For nonzero angular modes, a streamfunction gives

$$u_r = \frac{\psi_\theta}{r}, \qquad u_\theta = -\frac{\psi_s}{r}, \qquad \omega = -\frac{\psi_{ss} + \psi_{\theta\theta}}{r^2}. \tag{3.2}$$

The azimuthal mean is advanced separately as a real tangential velocity $v(s,t)$; its radial velocity is zero and its vorticity is $(v+v_s)/r$. This representation preserves all sine and cosine components of every retained nonzero mode and does not impose a reflection or zero-circulation constraint. The pressure is eliminated from the nonzero-mode equations by taking the curl; the mean tangential momentum equation supplies the remaining velocity block.

The initially stationary flow is driven by

$$U(t) = 1 - e^{-(t/0.8)^3}, \qquad V(t) = \begin{cases} 0.12\sin^4[\pi(t-4)/8], & 4 < t < 12, \\ 0, & \text{otherwise.} \end{cases} \tag{3.3}$$

The finite-duration transverse pulse breaks reflection symmetry. At the inner boundary, $\psi = \psi_s = 0$. At the outer boundary, the nonzero modes satisfy

$$\psi(L,\theta,t) = R[U(t)\sin\theta - V(t)\cos\theta], \qquad \omega(L,\theta,t) = 0. \tag{3.4}$$

The separate mean block satisfies $v(0,t) = 0$ and $v_s(L,t) + v(L,t) = 0$. Thus the outer circle does not have a full velocity Dirichlet condition. It is part of the definition of the finite-domain problem, rather than an exact transparent boundary for unbounded cylinder flow.

The prescribed boundary data are represented by a lifting

$$\begin{aligned} \psi_\ell(s,\theta,t) &= \ell(s)[U(t)\sin\theta - V(t)\cos\theta], \\ \ell(s) &= R[(1-\beta)z^2 + \beta z^3], \quad \beta = \frac{L^2-2}{4}. \end{aligned} \tag{3.5}$$

It obeys $\ell(0) = \ell_s(0) = 0$, $\ell(L) = R$, and $\ell_{ss}(L) = R$. The last identity enforces zero outer vorticity for the lifted $k = 1$ mode. With the convention $f = c_0 + 2\,\mathrm{Re}\sum_{k=1}^{K} c_k e^{ik\theta}$, the complex lifting amplitude is $g(t) = -(V(t) + iU(t))/2$. The same convention is used in reconstruction, nonlinear projection, and error scoring; conjugate negative modes represent the complete real field.

The finite outer radius is a material limitation. In the archived domain-size diagnostic, increasing R from 6 to 8 changed the unaligned vorticity by about 50.39% in the common $r \le 4$ region, and also changed the shedding frequency. The numerical tolerance below is therefore a tolerance for the specified $R = 6$ problem. Domain truncation and discretization error are not interchangeable.

3.2 Random and deterministic radial trial functions

Write the nonzero streamfunction coefficients as

$$\psi_k(s,t) = \sum_{j=1}^{d} F_j(s) x_{jk}(t) + \delta_{k1} g(t) \ell(s), \qquad 1 \le k \le K. \tag{3.6}$$

The functions F_j are fixed before time integration. No neural network weights are trained along the trajectory. The RFM designation refers to the randomly chosen radial feature space, followed by a deterministic Galerkin evolution of its coefficients [1–3].

For an odd input size $m = 2p + 1$, the raw random features are a constant and p sine–cosine pairs. Specifically,

$$\begin{gathered} b_j = \frac{62}{p}(j + \xi_j), \qquad \xi_j \sim \mathrm{Uniform}(0.2, 0.8), \\ q_{j,0}(z) = \cos[b_j(z - \tfrac{1}{2})], \quad q_{j,1}(z) = \sin[b_j(z - \tfrac{1}{2})]. \end{gathered} \tag{3.7}$$

where $j = 0, \ldots, p - 1$. The random draws use a fixed `NumPy default_rng(seed)`. The registered seed set is $7331, 7687, 8011, 8353, 9041$. The Legendre comparison replaces these raw functions by $P_j(2z - 1)$, $j = 0, \ldots, m - 1$, while retaining the same boundary transformation, enrichment, quadrature, and time discretization. The reported cases use no radial coordinate warping.

Boundary conditions are imposed analytically. For a raw function q, let $\chi(z) = 16z^2(1 - z)$ and $b(z) = z^2(1 - z)^2/2$, and define

$$\mathcal{B}q(z) = \chi(z) q(z) - b(z) \left.\frac{d^2(\chi q)}{dz^2}\right|_{z=1}. \tag{3.8}$$

Because $b(0) = b'(0) = b(1) = 0$ and $b''(1) = 1$, this function satisfies $F(0) = F_s(0) = F(L) = F_{ss}(L) = 0$. Consequently the homogeneous nonzero modes satisfy both no slip at the cylinder and zero outer vorticity for every k. Derivatives with respect to s include the factors L^{-1} and L^{-2}; the correction in (3.8) is expressed in z.

Both trial families receive the same six analytic outer-layer functions. For $\delta = L - s$, these are

$$E_\gamma(s) = z^2 \left[\delta + \left(\gamma + \frac{2}{L}\right)\delta^2\right] e^{-\gamma\delta}, \qquad \gamma \in \{16, 36, 80, 165, 340, 660\}. \tag{3.9}$$

Their constant and quadratic Taylor coefficients at $\delta = 0$ vanish, giving the same outer homogeneous conditions; the factor z^2 supplies the inner conditions. These functions resolve the outer-boundary layers of this finite-domain formulation. They are not a learned consequence of the random features.

The mean block uses a deterministic polynomial space in both families. Its raw functions are $z[(L + 2) - (L + 1)z]$ and $z(1 - z)^2 P_j(2z - 1)$, $j = 0, \ldots, m_0 - 2$, with $m_0 = 24$. The mean enrichment is $z[1 + (\gamma + 1 + 1/L)\delta]e^{-\gamma\delta}$. All satisfy the mean boundary conditions. The six enrichments give 30 retained mean functions in the final configurations. Randomness is therefore restricted to the nonzero-mode radial trial space.

3.3 Quadrature, rank selection, and the meaning of model size

Radial integration uses positive composite Gauss weights on $[0, L-0.12]$, $[L-0.12, L-0.018]$, and $[L-0.018, L]$. With q nodes and $b_q = \max(8, \lfloor q/6 \rfloor)$, the three allocations are $q-2b_q, b_q, b_q$. For the two principal Galerkin configurations, these are 44/11/11 and 52/12/12. The random trigonometric and exponential radial functions are not polynomials, so this quadrature is an approximation to the continuous weak form. Positivity statements below apply exactly to the quadrature-defined matrices; they are not quadrature-error estimates.

Let $\widetilde{F}, \widetilde{D}, \widetilde{D}_2$ be the values and first two s-derivatives of all raw, boundary-corrected and enriched functions at the radial nodes. With $W = \mathrm{diag}(w_i)$, form

$$\mathcal{A} = \mathrm{diag}(\sqrt{w_i}/r_i)(\widetilde{D}_2 - \widetilde{F}), \quad S = \mathrm{diag}(\|\mathcal{A}_{:j}\|_2), \quad \mathcal{A}S^{-1} = U\Sigma V^{\mathsf{T}}. \tag{3.10}$$

All retained raw columns must have nonzero norms. Keep singular values $\sigma_j > 10^{-10}\sigma_1$, and set

$$C = S^{-1}V_d\Sigma_d^{-1}, \qquad F = \widetilde{F}C, \quad D = \widetilde{D}C, \quad D_2 = \widetilde{D}_2C. \tag{3.11}$$

This is a discrete differential normalization with rank truncation, rather than an orthogonalization in the ordinary point-value norm. In exact arithmetic, $\mathrm{diag}(\sqrt{w_i}/r_i)(D_2 - F) = U_d$. It therefore normalizes the $k = 1$ dissipation Gram matrix to νI. This identity describes the existing construction; it is not a newly measured optimization or a guarantee for all angular modes. The mean block analogously normalizes $\mathrm{diag}(\sqrt{w_i})(\widetilde{V} + \widetilde{V}_s)$.

An input label such as RFM37 counts raw random features, not unknowns. Enrichment adds columns and truncation can remove them. The actual nonzero-mode rank is d, and the real coefficient count is $2Kd + d_0$, where $d_0 = 30$ is the mean rank. Fixed-rank contrasts and accuracy-matched contrasts answer different questions. In particular, a lower RFM input count is not evidence of a lower rank than a Legendre control. The seed and cutoff are part of the model identity and must remain fixed when comparing implementations.

3.4 The FD4 wall closure on the actual radial grid

The annular finite-difference baseline uses the logarithmic coordinate $s = \log(r/a)$, rather than a uniform physical-radius grid. Its registered outer-clustered grid is

$$s_j = L\sin\left(\frac{\pi j}{2n}\right), \qquad r_j = ae^{s_j}, \qquad j = 0, \ldots, n. \tag{3.12}$$

The qualified FD176/FD184 configurations and the two FD448 references use this same grid family. The canonical implementation is `work/rfm_v6/fd_full.py`, which calls `wall_second` in `work/rfm_v3/cylinder_fd.py`. It recomputes one-sided Hermite weights for the actual nonuniform nodes. The earlier `work/rfm_ns/cfd_reference_fourth.py` implements a Cartesian cavity and is not this annular baseline.

For the inner wall, set $h = s_1 - s_0$, $z_j = (s_j - s_0)/h$, and $d_w = \psi_s(0)$. The five value weights a_j and the slope weight b satisfy

$$\begin{aligned} \sum_{j=0}^{4} a_j z_j^m + b\,\delta_{m1} &= 2\delta_{m2}, \quad m = 0, \ldots, 5, \\ D_{s,w}^2\psi &= \frac{1}{h^2}\sum_{j=0}^{4} a_j\psi(s_j) + \frac{b}{h}d_w. \end{aligned} \tag{3.13}$$

Thus the closure reproduces polynomials through degree five. For smooth functions and bounded local node ratios it is fourth-order accurate. On the special uniform-s grid it reduces to

$$\begin{aligned} D_{s,w}^2\psi &= \frac{-\frac{415}{72}\psi(0) + 8\psi(h) - 3\psi(2h) + \frac{8}{9}\psi(3h) - \frac{1}{8}\psi(4h)}{h^2} - \frac{25}{6h}d_w, \\ D_{s,w}^2\psi - \psi_{ss}(0) &= -\frac{h^4}{15}\psi^{(6)}(0) + O(h^5). \end{aligned} \tag{3.14}$$

Omitting the wall-value term requires $\psi(0) = 0$; the constant coefficients must not be inserted unchanged on the nonuniform grid in Eq. (3.12). For the present nonzero angular modes, no slip gives $\psi_k(0) = \psi_{k,s}(0) = 0$, and the implemented wall vorticity is $\omega_k(0) = -a^{-2} D_{s,w}^2 \psi_k$. The angular-mean tangential-velocity block instead uses its separate first-derivative boundary reconstruction.

An additional spatial-only audit verifies the uniform coefficients and polynomial moments using exact rational arithmetic. The actual nonuniform rows at $n = 176, 184, 384, 448$ have a maximum absolute residual of approximately 2.84×10^{-14} for the six scaled moments in FP64; a degree-five test of the constructed FD184 wall-vorticity operator has absolute error 2.31×10^{-14}. These are stencil and implementation checks, not a new flow trajectory or a long-time convergence certificate.

4 Weak formulation and time-discrete algorithms

4.1 Nonzero-mode mass, dissipation, and nonlinear flux

For the velocity field defined above, incompressibility gives the vorticity equation in conservative logarithmic coordinates:

$$r^2 \partial_t \omega + \partial_s (r u_r \omega) + \partial_\theta (r u_\theta \omega) = \nu(\omega_{ss} + \omega_{\theta\theta}). \tag{4.1}$$

Test its k-th Fourier coefficient with a homogeneous radial basis function. One integration by parts in the mass term after inserting $r^2 \omega_k = -(\psi_{k,ss} - k^2 \psi_k)$ yields the bilinear form $\int (F_i' \psi_k' + k^2 F_i \psi_k)\, ds$. Integrating the viscous term twice gives a negative Gram form. The boundary terms vanish because $F_i = 0$ at both ends, $F_i'(0) = 0$, and $\omega_k(L) = 0$. Applying the prescribed radial quadrature gives

$$M_k = D^{\mathsf{T}} W D + k^2 F^{\mathsf{T}} W F, \tag{4.2}$$

$$K_k = \nu (D_2 - k^2 F)^{\mathsf{T}} \operatorname{diag}(w_i / r_i^2)(D_2 - k^2 F). \tag{4.3}$$

For every real coefficient vector x,

$$\begin{aligned} x^{\mathsf{T}} M_k x &= \|W^{1/2} D x\|_2^2 + k^2 \|W^{1/2} F x\|_2^2, \\ x^{\mathsf{T}} K_k x &= \nu \| \operatorname{diag}(\sqrt{w_i}/r_i)(D_2 - k^2 F) x\|_2^2. \end{aligned} \tag{4.4}$$

Both matrices are symmetric positive semidefinite in exact arithmetic. Positive definiteness of M_k additionally requires full column rank of the stacked weighted value and derivative samples on the retained space. The compact form (4.3) is algebraically equivalent to the expanded expression used by the archived implementation. Its positivity does not remove finite precision cancellation, assembly asymmetry, or a poorly conditioned coordinate map.

Let a_k and b_k be the Fourier coefficients of $u_r \omega$ and $u_\theta \omega$ evaluated at the radial nodes. Integration by parts in the two transport terms gives

$$f_k = D^{\mathsf{T}} \operatorname{diag}(w_i r_i) a_k - ikF^{\mathsf{T}} \operatorname{diag}(w_i r_i) b_k. \tag{4.5}$$

The products use the complete velocity, including the mean block and boundary lifting. They are evaluated in physical angle and projected back to the retained spectrum, coupling the otherwise independent linear mode blocks. This is the reason that making linear solves diagonal does not eliminate the nonlinear FFT work.

For $k = 1$, the lifting projections are

$$\begin{aligned} M_\ell &= D^{\mathsf{T}} W \ell_s + F^{\mathsf{T}} W \ell, \\ K_\ell &= \nu (D_2 - F)^{\mathsf{T}} \operatorname{diag}(w_i / r_i^2)(\ell_{ss} - \ell). \end{aligned} \tag{4.6}$$

They vanish from the equations of the other modes. Suppressing the mode index and allowing zero lifting vectors where appropriate, the semidiscrete system is

$$m'(t) + Kx(t) + g(t)K_\ell = f(t), \qquad m(t) = Mx(t) + g(t)M_\ell. \tag{4.7}$$

The differentiated quantity is the complete weak mass m. Treating it as Mx alone would omit the boundary acceleration. Several independent liftings can be included by replacing each gM_ℓ and gK_ℓ by the corresponding sum.

4.2 Why the mean block is a separate equation

Angular averaging of tangential momentum gives

$$\partial_t v = -\langle u_r \omega \rangle_\theta + \frac{\nu}{r^2}(v_{ss} - v). \tag{4.8}$$

Let V and V_s now denote the sampled mean basis and its derivative, rather than the transverse pulse in (3.3). The mean matrices and source are

$$\begin{aligned} M_0 &= V^{\mathsf{T}} \operatorname{diag}(w_i r_i^2) V, \qquad K_0 = \nu (V + V_s)^{\mathsf{T}} W (V + V_s), \\ f_0 &= -V^{\mathsf{T}} \operatorname{diag}(w_i r_i^2) a_0. \end{aligned} \tag{4.9}$$

To check the dissipation form, expand $\int (V + V_s)(v + v_s)\, ds$: its cross terms combine to $[Vv]_0^L$, which is exactly the boundary contribution obtained from $-\int V(v_{ss} - v)\, ds$ using $v_s(L) = -v(L)$ and $v(0) = 0$. There is no direct boundary lifting in this block. The negative radial-flux sign in (4.9) follows from tangential momentum. Setting $k = 0$ in (4.2)–(4.5) would produce a different representation and is not the implemented mean equation.

4.3 Constant-step integration and a single bridge

The baseline treats diffusion implicitly and the nonlinear weak flux by two-point extrapolation, using SBDF2 [11, 12]. After the implicit Euler start, a constant step h solves

$$\begin{aligned} (\tfrac{3}{2}M + hK)x_{n+1} = 2m_n - \tfrac{1}{2}m_{n-1} + h(2f_n - f_{n-1}) \\ - g_{n+1}(\tfrac{3}{2}M_\ell + hK_\ell). \end{aligned} \tag{4.10}$$

The initialization uses the stationary initial state and the prescribed smooth startup. Euler startup does not establish second-order convergence by itself; its consistency with the initial-value problem must be considered together with the following multistep evolution. In the implemented initial state the explicit nonlinear source is zero.

The optimized schedule uses a constant step H until $t_0 = 8$ and a larger constant step h afterward. Only the first update across this interface has unequal consecutive intervals. Write $t_- = t_0 - H$, $t_+ = t_0 + h$, and $\rho = h/H > 0$. Define

$$a_0 = \frac{1 + 2\rho}{1 + \rho}, \qquad b_0 = 1 + \rho, \qquad b_1 = \frac{\rho^2}{1 + \rho}. \tag{4.11}$$

The bridge is

$$\begin{aligned} (a_0 M + hK)x_+ = b_0 m_0 - b_1 m_- + h[(1 + \rho)f_0 - \rho f_-] \\ - g_+(a_0 M_\ell + hK_\ell). \end{aligned} \tag{4.12}$$

In this subsection m_0 and f_0 refer to time t_0, not to the azimuthal mean. Historical masses already contain their historical lifting values; the future lifting is subtracted once. At $\rho = 1$, (4.12) reduces exactly to (4.10).

Proposition 1 (local consistency of the lifted bridge). Assume $m \in C^4$ and $f \in C^3$ along a smooth exact semidiscrete solution in a neighborhood of the bridge. Assume also that ρ lies in a fixed compact subset of $(0, \infty)$. For $\delta = \max(H, h)$, the derivative-form residual of (4.12) is $O(\delta^2)$, and its step-scaled equation residual is $O(h\delta^2)$.

Proof. Differentiate the quadratic interpolant through the three masses at t_+. Its derivative is $Dm_+ = (a_0 m_+ - b_0 m_0 + b_1 m_-)/h$. The linear extrapolant of the two fluxes has value $(1+\rho)f_0 - \rho f_-$ at the same endpoint. Taylor expansion about t_+ gives

$$Dm_+ - m'(t_+) = -\frac{h(h+H)}{6} m'''(t_+) + O(\delta^3), \tag{4.13}$$

$$(1+\rho)f_0 - \rho f_- - f(t_+) = -\frac{h(h+H)}{2} f''(t_+) + O(\delta^3). \tag{4.14}$$

Substitute these in (4.7), evaluate diffusion and lifting at t_+, and collect the two residuals. Substitution of $m_+ = Mx_+ + g_+ M_\ell$ gives the stated bridge. □

This proposition is local consistency, not unconditional stability or long-time convergence of the nonlinear Navier–Stokes solution. Those require additional regularity, startup, perturbation, and stability estimates. The finite-duration pulse is piecewise smooth; the displayed Taylor argument is local to the bridge at $t = 8$, which is away from the pulse endpoints. For the scalar zero equation $y' = 0$, the bridge differences satisfy

$$y_+ - y_0 = \frac{\rho^2}{1+2\rho}(y_0 - y_-). \tag{4.15}$$

The two registered ratios 8/7 and 19/17 give 64/161 and 361/935; the constant segments have factor 1/3. Thus the homogeneous differences are bounded on this particular grid. Explicit advection and mode coupling are absent from this check, so it does not supply a flow stability theorem.

The work reduction follows from the schedule rather than from the trial family. At $T = 72$, changing the rate from 128 to 112 at $t = 8$ changes the step count from 9216 to 8192. Changing 152 to 136 changes 10944 to 9920. The initial nonlinear source is identically zero and the Euler state is constructed before the repeated updates. Nonlinear kernels evaluate the states at $t_1, \ldots, t_{N_{\text{step}}-1}$; the terminal state needs no subsequent evaluation. The respective recorded evaluation counts are $9215 \to 8191$ and $10943 \to 9919$. This count does not predict the complete solve time, which also includes additional phase preparation and startup.

4.4 Reuse of fixed spatial preparation

Proposition 2 (exact-arithmetic reuse). Fix the geometry, viscosity, feature family, random seed, enrichments, quadrature, lifting, column scaling, and singular-value cutoff. If the basis transformation is formed before any step-dependent matrix and does not depend on the time step, then the retained bases, ranks, M, K, M_ℓ, K_ℓ, and uncontracted nonlinear test operators can be prepared once for all time phases without changing the discrete update in exact arithmetic.

Proof. Equations (3.10)–(3.11) and (4.2)–(4.9) contain no time-step parameter. Phase dependence enters only through $a_j M + h_j K$, its application or factorization, and the step-weighted right-hand side. Reusing the same inputs to these operations therefore gives the same linear equations. □

The existing assembly path accepts $A_{\text{eff}} = (3/2)M + h_{\text{eff}} K$. For the bridge set $c = (3/2)/a_0$ and $h_{\text{eff}} = ch$. Then

$$A_{\text{eff}} = c(a_0 M + hK), \qquad A_{\text{eff}}^{-1}(cb) = (a_0 M + hK)^{-1} b. \tag{4.16}$$

Historical mass coefficients are multiplied by c, while a flux package already containing h_{eff} must not receive that factor a second time. Previously precontracted nonlinear histories depend on the old inverse matrix and must be remapped from their original flux spectra at the interface. Reusing those old contracted vectors directly would change the algorithm.

Exact-arithmetic reuse does not imply bitwise equality across implementations. The archive verifies byte-for-byte equality of 936 arrays in 24 fixed comparisons, retaining the original floating-point expression grouping and environment. Changing the reduction order, library, spatial basis, lifting, or adaptation policy requires a new check. Shared preparation removes repeated spatial work but leaves the phase-specific factorizations, operator packages, transfers, and execution graphs; it is not by itself a timing result.

5 Common phase coordinates and the limits of algebraic improvement

5.1 Simultaneous congruence for one angular mode

For one fixed mode, let the phase matrices be $A_j = a_j M + h_j K$, with $a_j, h_j > 0$. The two constant segments and the bridge have this form. We use a different symbol for the reference matrix, H_*, to distinguish it from the early time step H. Assume that M, K are real symmetric positive semidefinite and $H_* = a_* M + h_* K$ is positive definite for $a_*, h_* > 0$. Equivalently, M and K have no common nonzero null vector. The standard symmetric-definite generalized eigenproblem [13] provides the following representation.

Proposition 3 (common coordinates). There exists a real invertible P such that $P^{\mathsf{T}} H_* P = I$, $P^{\mathsf{T}} M P = \operatorname{diag}(\lambda_i)$, and, with $\theta_i = a_* \lambda_i$,

$$\begin{aligned} 0 \le \theta_i \le 1, \qquad & P^{\mathsf{T}} A_j P = \operatorname{diag}(d_{ji}), \\ d_{ji} = \frac{a_j}{a_*}\theta_i + \frac{h_j}{h_*}(1-\theta_i) > 0. & \end{aligned} \tag{5.1}$$

Proof. Write $H_* = LL^{\mathsf{T}}$. The symmetric positive semidefinite matrix $L^{-1} M L^{-\mathsf{T}}$ has an orthogonal eigenbasis Q. For $P = L^{-\mathsf{T}} Q$, its eigenvalues give $P^{\mathsf{T}} M P = \operatorname{diag}(\lambda_i)$ and $P^{\mathsf{T}} H_* P = I$. Further, $h_* P^{\mathsf{T}} K P = I - a_* \operatorname{diag}(\lambda_i)$ is positive semidefinite, so $0 \le a_* \lambda_i \le 1$. Substitute the linear combination defining A_j to obtain (5.1). Every diagonal value is a convex combination of two positive numbers. □

Consequently,

$$A_j^{-1} b = P \operatorname{diag}(d_{ji}^{-1}) P^{\mathsf{T}} b. \tag{5.2}$$

This is a congruence transformation, rather than a claim that the original-coordinate matrices commute. The construction also applies to the separate mean matrices under the same hypotheses. Although an Euler matrix has positive coefficients and fits the algebra, the archived common-phase prototype retains the original Euler startup; that phase was not independently promoted by the prototype tests.

5.2 A geometric anchor for a worst-case spectral bound

Set $r_j = h_j / a_j$, $\eta = h_* / a_*$, $r_{\min} = \min_j r_j$, and $r_{\max} = \max_j r_j$. After extracting a_j / a_* from d_{ji}, its remaining factor is $\theta_i + (r_j/\eta)(1 - \theta_i)$, which lies between 1 and r_j/η. Therefore

$$\kappa_2(P^{\mathsf{T}} A_j P) \le B_j(\eta) = \max\{r_j/\eta, \eta/r_j\}. \tag{5.3}$$

For a finite set of positive phases,

$$\eta_* = \sqrt{r_{\min} r_{\max}}, \qquad \min_{\eta > 0} \max_j B_j(\eta) = \sqrt{r_{\max}/r_{\min}}. \tag{5.4}$$

Indeed, the worst bound is $\max\{r_{\max}/\eta, \eta/r_{\min}\}$. The two positive terms have fixed product $r_{\max}/r_{\min}$; their maximum is smallest when they are equal, proving (5.4).

The distinction between the bound and the actual spectrum is essential. If the θ_i do not approach both endpoints, the interval estimate is loose. The anchor minimizes that estimate, not necessarily the actual maximum condition number for the given matrices. Rescaling a_*, h_* together does not change this bound but does change diagonal magnitudes, so the rule does not imply that all denominators are close to one. It also does not improve $\kappa_2(A_j)$ in the original coordinates or bound the errors of applying P and P^{T}.

The geometric-anchor statement is an algebraic derivation, separately reviewed, and was not a timed candidate. The measured setup overhead of the earlier common-phase prototype cannot be assigned to this untested choice of anchor. Neither the derivation nor the prototype constitutes a random-feature approximation theorem or a stability theorem for the time integrator.

5.3 Why common coordinates need not yield a faster solver

Across angular modes, $M_k = M^{(0)} + k^2 M^{(1)}$ and $K_k = K^{(0)} + k^2 K^{(1)} + k^4 K^{(2)}$. Polynomial dependence on k^2 does not imply a common eigenbasis. In a fixed whitening metric, real symmetric coefficient matrices admit a common orthogonal diagonalization precisely when they commute pairwise. The archived normalized commutators, about 0.051–0.076, do not support replacing all modes by one exact common transform.

Even within one mode, the storage contract matters. Suppose the device expects precontracted dense operators $A_j^{-1}B$, where B contains mass and nonlinear projection columns. Keeping this interface requires $P\,\mathrm{diag}(d_j^{-1})P^{\mathsf{T}}B$ for every phase. For rank d, projection width proportional to q, and p_{ph} phases, this entails an $O(d^3)$ eigendecomposition and approximately $p_{\mathrm{ph}}+1$ $O(d^2 q)$ dense multiplications. Direct phase preparation uses p_{ph} factorizations and phase-wise applications instead. The common representation still writes all phase packages. Operation counts, library throughput, conditioning, and memory layout therefore have to be evaluated together.

A different design could keep the time-dependent coefficients in the P coordinates and use diagonal phase updates throughout. It would also require mode-dependent transformed reconstruction, lifting, nonlinear projection, and observation operators. The current trajectory evidence does not validate that design. It remains a specific engineering opportunity, rather than an inferred speedup from (5.2).

5.4 Separate approximation, integration, and implementation errors

Four levels are relevant: the continuous finite-domain solution; a quadrature-defined spatial semidiscretization; its exact-arithmetic time-discrete solution; and the executed floating-point solution. They distinguish domain error, trial-space and quadrature error, time-discretization error, and implementation error. On a common physical reconstruction grid, the triangle inequality gives an additive decomposition in a common absolute norm. Relative error bounds require explicit denominator conversions; two percentages with different reference fields cannot simply be added.

The observed field gates in this paper compare numerical trajectories to two resolved numerical references. The implementation gate compares GPU and CPU realizations of the same discrete method. Neither is a rigorous upper bound on error against the unknown continuous solution. In addition, a small velocity error does not control its spatial derivatives: (3.2) differentiates the streamfunction twice for vorticity. Vorticity and near-field gates must therefore be imposed independently.

For completeness, a conditional perturbation estimate explains why small operator residuals still require trajectory checks. Consider two executions of the same step and lifting, with

coefficient errors e_n and e_{n-1}, and an additive algebraic residual r_{n+1} in the step-scaled linear equation. If the nonlinear map is Lipschitz with constant L_f on a region containing both trajectories, then a compatible coefficient norm gives

$$\begin{aligned}\|e_{n+1}\| \le \|A_n^{-1}\|\big[&(b_0\|M\| + h(1+\rho)L_f)\|e_n\| \\ &+ (b_1\|M\| + h\rho L_f)\|e_{n-1}\| + \|r_{n+1}\|\big].\end{aligned} \tag{5.5}$$

Subtracting the two updates and applying the norm and local Lipschitz inequality proves this estimate. It is only error bookkeeping for the same finite-dimensional scheme; its constants may be very pessimistic, and it does not establish a uniform long-time bound. It explains why agreement of an isolated FFT, small linear residuals, or a favorable transformed condition number cannot replace the physical accuracy checks at the full integration horizon.

6 Low-level operators and computational cost

6.1 An operator boundary that preserves the discretization

The nonlinear stage begins after radial reconstruction has produced the angular half-spectra of the physical fields. It ends before radial projection or construction of the finite-difference right-hand side. This boundary is essential: a replacement kernel is required to compute the same dealiased products from the same reconstructed coefficients. Reducing the number of time samples, changing the quadrature, or replacing a weak flux form by a different discrete convection form would alter the method and requires a separate accuracy study. None of those changes is implicit in the fusion described here.

Let q denote the number of radial rows for this stage, N the angular transform length, and $H_N = \lfloor N/2 \rfloor + 1$ the half-spectrum width. The layout is field-major, with address $((cq+i)H_N+k)$ for field c, row i, and nonnegative mode k. Each element is a complex FP64 pair. The Galerkin path reads three planes for u_r, u_θ, ω and returns two planes for $u_r\omega, u_\theta\omega$. The FD path reads four planes and returns one for the sum of two derivative products; its radial metric factors are supplied by the existing reconstruction and right-hand-side conventions. The FD independent mean forcing is supplied separately and retains the original ascending modal sum. Both variants copy the previous raw flux into the prescribed SBDF2 history buffer before overwriting the current buffer.

The reference implementation makes one batched inverse cuFFT call, a product-kernel call, and one batched forward cuFFT call. A batch contains many field–row transforms. Counting three API-level operations is therefore different from counting three individual transforms, and a cuFFT call may itself launch several device kernels[14]. Galerkin requires three real inverse and two real forward transforms per row; FD requires four inverse and one forward. Both have five logical field transforms. The optimization preserves this logical work and changes its realization.

6.2 Packing identities and normalization

Define the unnormalized forward and inverse transforms by

$$(\mathcal{F}_N v)_k = \sum_{j=0}^{N-1} v_j e^{-2\pi \mathrm{i} jk/N}, \qquad (\mathcal{B}_N A)_j = \sum_{k=0}^{N-1} A_k e^{2\pi \mathrm{i} jk/N}. \tag{6.1}$$

Thus $\mathcal{B}_N\mathcal{F}_N = NI$. The CUDA replacement follows the cuFFT convention[14]; it does not insert an additional $1/N$. In the solver, the corresponding normalization already belongs to the following projection or FD right-hand side. Applying it in both locations would change the nonlinear equation, even if a standalone round-trip transform appeared correct.

For two separately Hermitian spectra A, B, linearity gives

$$\mathcal{B}_N(A + \mathrm{i}B) = a + \mathrm{i}b, \qquad a = \mathcal{B}_N A \in \mathbb{R}^N, \quad b = \mathcal{B}_N B \in \mathbb{R}^N. \tag{6.2}$$

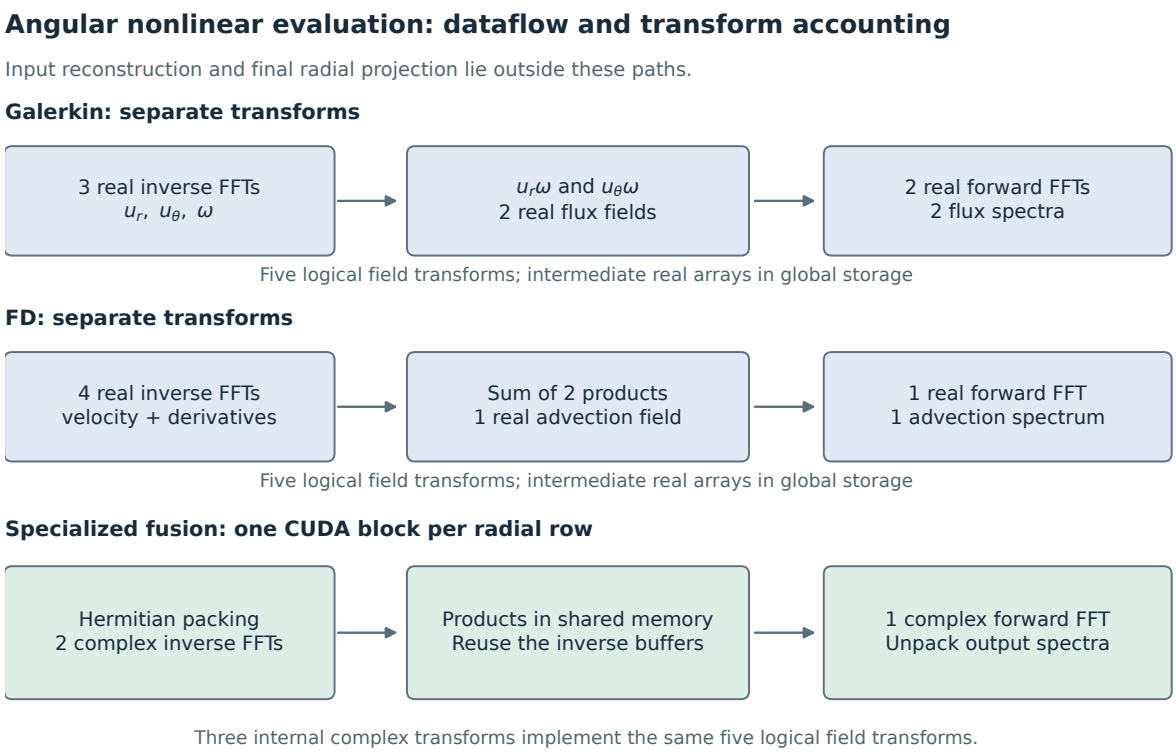


Figure 1: The nonlinear operator boundary. The two discretizations retain their respective field products. Fusion combines the angular transforms, physical products, and packing in shared memory; radial reconstruction and projection lie outside the illustrated replacement. Five logical field transforms remain in both variants.

The negative modes must be reconstructed as $\overline{A_k} + \mathrm{i}\overline{B_k}$. Conjugating the packed positive mode as a whole instead produces $\overline{A_k} - \mathrm{i}\overline{B_k}$ and is incorrect. The packed sequence is generally not Hermitian. At zero frequency, and at Nyquist frequency for even N, each original real-field coefficient has zero imaginary part.

After forming two real products g, h, let $Z = \mathcal{F}_N(g + \mathrm{i}h)$. Their separate spectra follow from

$$G_k = \frac{Z_k + \overline{Z_{-k}}}{2}, \qquad H_k = \frac{Z_k - \overline{Z_{-k}}}{2\mathrm{i}}. \tag{6.3}$$

The Galerkin implementation uses one packed inverse for the two velocities, one inverse for vorticity, and one packed forward for the two fluxes. The FD implementation packs the two velocities and the two derivatives separately, forms their dot product, and performs one forward transform. Consequently both use three internal complex transforms. This accounting is an implementation identity, not evidence of a factor 5/3 speedup: real cuFFT transforms exploit Hermitian redundancy already, and the packing, permutation, synchronization, and instruction costs must also be paid.

For a retained band $|k| \le K$, quadratic products have support at most $2K$. An aliased contribution into a retained mode would require $|k + \ell N| \le 2K$ for some nonzero integer ℓ. Since $|k + \ell N| \ge N - K$, the strict condition $N > 3K$ excludes such a contribution. This is the standard retained-band dealiasing argument[15]. It does not imply that modes above K vanish in the product, nor does it remove radial quadrature error. The replacement preserves the original padded angular grid and truncation.

6.3 Block mapping, synchronization, and memory traffic

One 128-thread block handles one radial row. Two shared arrays of N complex doubles hold the packed inverse inputs and physical fields. The specialized lengths and factorizations are $200 = 2^3 5^2$, $225 = 3^2 5^2$, and $243 = 3^5$. Their shared storage is respectively 6400, 7200, and 7776 bytes per block. These are the supported kernel instantiations; the code is not a general arbitrary-length FFT service.

Let the ordered radices be $r_0, \ldots, r_{s-1}$, and express an input position using mixed-radix digits d_j. The load permutation is

$$p(i) = \sum_{j=0}^{s-1} d_j \frac{N}{\prod_{\ell=0}^{j} r_\ell}, \qquad i = \sum_{j=0}^{s-1} d_j \prod_{\ell=0}^{j-1} r_\ell. \tag{6.4}$$

At a stage with previous-radix product m and current radix r, disjoint butterflies combine positions separated by m, after the appropriate twiddle multiplication. Each thread visits butterflies in strides of 128. A block barrier separates consecutive stages. This is a classical mixed-radix FFT construction; the project contribution is its specialized mapping within this operator boundary, not a new FFT factorization.

The complete sequence is: load the two packed inverse planes in permuted order; execute the inverse stages; retain each thread's one or two physical products in registers; synchronize all reads; overwrite the first shared plane with the permuted products; execute the forward stages; unpack and store the output half-spectra. The register retention and barrier before overwrite are necessary because another thread may still need the old inverse output. Omitting that barrier introduces a read/write race despite using a single block. Including initial-load and overwrite barriers, the number is $2s + 3$, or 13, 11, and 13 for the supported lengths.

The separate implementation materializes inverse physical arrays and forward product arrays in global memory. Counting writes and reads of these named arrays gives a nominal source-level reduction of $80Nq$ bytes per nonlinear evaluation in either variant. For Galerkin, the count is $(3 + 3 + 2 + 2)Nq$ doubles; for FD it is $(4 + 4 + 1 + 1)Nq$ doubles. This is an access model for the named real arrays, not a bound on total eliminated device traffic. It excludes cache reuse, compiler behavior, library workspaces, and traffic still required for spectral input, flux output, and flux history. It must not be reported as measured DRAM traffic or converted directly into a bandwidth speedup.

Fusion also increases shared-memory use, register lifetimes, and per-block work. The specialized row count may expose too few blocks to saturate a GPU, while consumer hardware may have limited FP64 throughput. Therefore reduced temporary storage does not by itself establish whether the resulting kernel is bandwidth-bound, instruction-bound, or latency-bound. A roofline or occupancy conclusion would require hardware counters and controlled measurements not collected in this study.

6.4 Floating-point limits and a guarded standalone extraction

Equations (6.2)–(6.3) hold in exact arithmetic. In floating-point arithmetic, a packed channel of characteristic magnitude A can introduce an absolute perturbation on the scale of ϵA into a second channel of magnitude B. The corresponding relative scale is $\epsilon A/B$, before considering transform constants or cancellation. This is an explanatory scale estimate, not an error bound for this kernel. It explains why a combined-vector norm can hide a poor relative error in a weak output. Near-cancelling product sums require an additional absolute error scaled by the magnitudes of the uncancelled terms.

The revision extracts the original CUDA file and its existing CPU model without modifying their bytes, and adds a solver-independent harness with new NumPy and direct-convolution oracles. The input contract checks supported lengths, FP64 contiguous layouts, real endpoints,

zero padding, finite data, and valid retained bandwidth. As a new engineering guard, the adapter rejects a pair of nonzero packed input spectra if the smaller per-row maximum magnitude is less than 10^{-8} times the larger. The threshold is a conservative dispatch policy, not a sufficient condition for a relative-error theorem. Exactly zero channels instead require an absolute-error check. A production integration would need a verified unpacked fallback and a low-cost way to obtain the relevant scales; neither is claimed here. The frozen NS kernels and their qualification records are unchanged.

The harness was run on the local GPU for 36 fixed-state cases: three lengths, two variants, and six fixture classes per variant. The sixth class is a zero velocity channel for Galerkin and additive cancellation for FD, giving seven distinct labels across the two variants. It uses both an independent NumPy FFT oracle and an FFT-free retained-band convolution oracle; it also checks flux history and FD mean-buffer parity. For each output channel, the relative metric is the infinity-norm ratio $\max_{i,k} |Y_{ik} - Y_{ik}^{\mathrm{ref}}| / \max_{i,k} |Y_{ik}^{\mathrm{ref}}|$, over all radial rows and half-spectrum modes, not a pointwise relative error. Ordinary cases attained a maximum channel-relative error of 1.18×10^{-15}. The absolute diagnostic uses one common scale across channels: the maximum, over output channels and rows, of N times the sum of the products of the separate physical-field maximum magnitudes in each flux expression. Across all fixtures, the largest absolute error divided by this uncancelled scale was 3.84×10^{-16}. Because this common scale can hide errors in a weak channel, it must be read together with the per-channel relative metrics. Deliberately reducing one velocity channel by 10^{-10}, after confirming rejection by the guard and explicitly overriding it for diagnosis, produced a maximum channel-relative error of 4.98×10^{-6}. These rounded-up figures describe new synthetic verification, distinct from the earlier archived operator measurements. No new timing or NS trajectory was collected for this extraction. It is distributed as a local contribution candidate in the supplementary source package.

6.5 Observation reduction and CPU kernels

The probe operator evaluates six weighted sums, $p_j = w_j^{\top} x + \ell_j(t)$. The old implementation assigned each sum to one thread. The replacement assigns a complete warp to each probe: lane ℓ accumulates indices $\ell + 32m$, then five shuffle reductions combine lane values[17]. A block of 192 threads therefore covers all six probes. The arithmetic work remains linear in the frame length; the per-probe serial depth is reduced from roughly L to $\lceil L/32 \rceil + 5$, ignoring instruction and memory costs. Snapshot copies, selected state pointers, and synchronization before advancing the state are preserved. Reordering additions changes rounding, so bitwise state preservation and numerical probe agreement are separate assertions. This common observation improvement is applied to both Galerkin and FD comparisons.

There are also project-local C implementations below the Python driver. The Galerkin contraction kernel places the angular mode in the contiguous inner loop and combines the same precontracted mass and flux projections. The buffered FD kernel applies already assembled narrow radial stencils, reuses working arrays, and forms physical products in place. A separate band solver applies cached real LAPACK factor data to interleaved complex right-hand sides. The FD build specifies optimization and AVX2 support while disabling fused multiply-add contraction; that compiler option alone does not certify SIMD utilization or a speedup. SciPy still provides selected FFTs or factorization services in these CPU variants. The project does not replace their algorithms or patch their source.

These implementations demonstrate genuine low-level changes, but their existence is not an isolated end-to-end timing result. Likewise, replacing the Python orchestration language with Rust would not directly reduce the device FFT work, retained rank, or compiled-kernel initialization. A language migration would be justified only by a measured host bottleneck or a reusable interface requirement. The current attribution distinguishes loop/layout changes from module import, compilation caching, and CUDA graph scheduling.

6.6 Costs, amortization, and upstream scope

For attribution, write the total cost as

$$T_{\text{total}} = T_{\text{start}} + T_{\text{prepare}} + N_{\text{nl}} C_{\text{nl}} + N_{\text{step}} C_{\text{lin}} + N_{\text{obs}} C_{\text{obs}} + T_{\text{finish}}. \tag{6.5}$$

The costs denote phase-weighted effective quantities, not an assertion that every step costs the same. Here C_{lin} excludes the nonlinear and observation terms already counted. Time-grid optimization changes N_{nl} and N_{step}; fusion changes C_{nl}; warp reduction changes C_{obs}; and preparation reuse or caching changes other terms. Their savings cannot be added across unrelated measurements or counted twice.

If a fraction f of an otherwise fixed execution is accelerated by a local factor s, the corresponding ideal bound is $[1 - f + f/s]^{-1}$. An additional initialization cost makes the gain smaller. More explicitly, let S_b, S_f be setup costs and B_b, B_f the costs of one otherwise matched repeated workload. Under genuine reuse of setup, the fused path wins after m repetitions only if

$$m(B_b - B_f) > S_f - S_b. \tag{6.6}$$

If each new process rebuilds or reloads the relevant state, setup is not a once-only term and this amortization model must be changed. The fixed-state event measurements and short-trajectory results in the next section illustrate precisely why the distinction matters. An explicit PTX cache also does not eliminate all driver JIT, loading, context, planning, or process-start costs.

The complete-solve clock includes construction, compilation or loading, transfers, FFT preparation, graph capture, advancement, observation, synchronization, download, and resource release. The new-process clock additionally includes interpreter startup, imports, and the prescribed supervision overhead. Neither clock can be replaced by isolated kernel events. Numerical qualification may use concurrent independent jobs, but latency comparison requires serialized GPU use and monitored external activity. Aggregate CPU utilization is not an accuracy-constrained performance metric.

Finally, the source audit found project-local application kernels and library calls, not an identified modification to borrowed cuFFT, cuFFTDx, SciPy, or LAPACK source. NVIDIA already provides FFT fusion and convolution examples, including comparisons with a cuFFT pipeline and optional LTO callbacks[16]. The plausible upstream contribution is therefore a new multifield FP64 inverse-transform–cross-product–forward-transform example, not a claim to have invented FFT fusion or repaired cuFFT. A useful proposal still requires an equivalently fused cuFFTDx implementation, a batched cuFFT baseline, matched precision and padding, first-use and repeated-use costs, and a clean build outside this machine. The local extraction has not satisfied those performance and portability requirements, has not established a new license grant or contribution certification, and has not been submitted as a PR[19].

6.7 Maintenance of the early steady-solver interface

The dynamic replacement cited by the reviewer belongs to an early steady-cavity control: its wrapper temporarily replaced `rfm_v2_solver.trial_basis`, outside the final annular solver path. We preserve the archived source hashes and provide a separate maintained package with an explicit, solve-local basis factory. Omitting this argument preserves the default behavior; the mapped-Legendre wrapper no longer changes module globals. All seven CPU tests pass. Eight small configurations agree with the archived functions in coefficients, Stokes matrices, and velocity, vorticity and other fields at 25 independent points, using relative and absolute tolerances of 10^{-12}. Additional tests force concurrent factory overlap and check default calls after factory and assembly exceptions. These establish interface compatibility and state isolation for this change; they provide no new annular accuracy certificate, GPU speed result, or thread-safety guarantee for arbitrary user-supplied factories.

7 Experimental design and attribution protocol

The experiments answer three different questions: whether a discretization reaches the prescribed physical accuracy, whether an implementation preserves that discretization, and whether a qualified implementation reduces complete solution time. These questions require different controls. A successful trajectory is evidence for numerical feasibility; a fast isolated operator is evidence about that operator; a complete paired timing campaign is required to establish a practical speed advantage. We retain these distinctions when reporting both successful and unsuccessful candidates.

7.1 Problem, references, and independent reconstruction

All main comparisons concern the finite annulus with $a = 0.5$, $R = 6$, $\nu = 0.01$, and $T = 72$. The final incident speed and cylinder diameter are both unity, giving $\mathrm{Re} = 100$. The transverse pulse ends at $t = 12$, whereas the observations extend to $t = 72$. No reflection constraint is imposed, and neither the fields nor the probe series are shifted in time or aligned by wake phase. This is necessary because accumulated phase error is part of the approximation error for an initial-boundary-value problem.

The two frozen numerical references, denoted `angular_fine` and `time_fine` in the archive, resolve angular and temporal refinement separately within a finite-difference refinement chain. The chain increases the radial resolution from 384 to 448 intervals, angular truncation from 128 to 144, and step rate from 384 to 512. They are distinct numerical references, not two independently implemented continuum-solution certificates. Their combined late-time discrepancy is 0.0290144%. This is below the empirical reference allowance of 0.125%, one quarter of the candidate tolerance. It is above the 0.025% allowance that the same rule would assign to a 0.1% candidate tolerance. Consequently the current reference evidence supports the reported 0.5% protocol, without establishing a certified continuum error or a tighter accuracy target.

Evaluation is performed after reconstructing every method on the same physical grids. For reconstruction grid G, checkpoint t, and its physical weights, define

$$E_u(G,t) = \frac{\|\boldsymbol{u} - \boldsymbol{u}_{\mathrm{ref}}\|_G}{\|\boldsymbol{u}_{\mathrm{ref}}\|_G}, \qquad E_\omega(G,t) = \frac{\|\omega - \omega_{\mathrm{ref}}\|_G}{\|\omega_{\mathrm{ref}}\|_G}. \tag{7.1}$$

The measure is $r^2\, ds\, d\theta$, and the velocity norm combines both polar components. Each quantity is checked over the full annulus and over $r \le 2.5$. The reconstruction grids use 123, 129, and 144 radial points and respectively 433, 449, and 457 shifted angular points. They differ from the solver quadrature and FFT grids. Their construction and the 19 physical checkpoint times are listed in Appendix A. Taking the maximum across grids reduces dependence on an individual sampling pattern, but remains a finite set of discrete tests.

There are 19 reference-scored physical checkpoints from $t = 0.5$ to 72. A piecewise schedule saves two additional states immediately after the bridge for implementation comparison; the archive therefore distinguishes 19 physical checkpoints from 21 saved implementation checkpoints. Six probe histories, transverse velocity and vorticity at three downstream positions, contain 577 equally spaced times, including the zero initial value. The implementation tolerance against the CPU trajectory of the same discretization is 10^{-5} for the field and complete probe-series comparisons. The wake checker additionally verifies sustained alternating signs. Neither a plausible shedding frequency nor a visually similar wake substitutes for the field-error tests.

7.2 A safeguard against compensating errors

For a piecewise schedule, the constant-step control uses the same space, seed, geometry, and quadrature. Let v be this control, w the piecewise candidate, and z a reference, all evaluated in

Table 2: Comparison matrix and the effect each contrast can identify. All physical comparisons retain the same boundary-value problem and error criteria.

Contrast	Held fixed	Quantity allowed to change
RFM / Legendre at equal rank	K, q, retained rank, schedule, shared kernels and runtime policy	Trial functions and resulting approximation error
Accuracy-matched spaces	Reference, field and probe tests, runtime policy	Rank and qualified schedule; actual rank reported per seed
Constant / piecewise	Same trial space, seed, K, q and observation protocol	Step schedule and associated phase preparation
cuFFT / fused operator	Input state, precision, retained spectrum, normalization and output definition	Transform implementation and temporary storage
Serial / warp observation	State evolution and probe weights	Floating-point summation order and observation cost
First use / cache hit	Solver, artifacts, accuracy criteria and fresh-process entry	Explicit compilation-artifact availability
Galerkin / FD	Physical problem, accuracy target, observation and runtime policies	Spatial discretization and its qualified resolution

the same weighted norm. Define $b = \|v - z\|_G / \|z\|_G$ and $d = \|w - v\|_G / \|v\|_G$. Then

$$\frac{\|w - z\|_G}{\|z\|_G} \leq b + d\frac{\|v\|_G}{\|z\|_G} \leq b + (1 + b)d. \tag{7.2}$$

Both the direct error and $b + (1 + b)d$ must be at most 0.005. Using $b + d$ would incorrectly ignore the different denominators. The bound is evaluated for each prescribed field, region, grid, checkpoint, and reference, before taking maxima. It prevents a candidate from qualifying solely because the schedule error cancels part of its constant-step error. It is a rigorous relation between these discrete fields, not an a posteriori bound against the exact PDE solution.

This extra test can reject a candidate whose direct error passes. For high-resolution Legendre34, the piecewise direct field score is approximately 0.481690%, but the triangle safeguard is 0.509600%. Its constant-step variant remains eligible. Keeping that cheaper deterministic alternative is essential: applying a common accuracy target does not require every method to use an identical schedule when the question is minimum cost at that target.

7.3 Controlled comparisons and computational environment

Table 2 separates the effects being compared. A same-rank comparison controls algebraic dimension but does not prove equal approximation power for all flows. A comparison of the least expensive qualifying configurations instead permits different ranks and schedules. Both are useful, and their conclusions must be labeled accordingly. The finite-difference baseline uses fourth-order radial spatial differences and the same second-order SBDF2 time family; "fourth order" does not describe its time discretization.

The recorded experimental machine and software are summarized in Table 3. The numerical backend uses FP64 CUDA kernels and cuFFT through a native interface. The CPU reference and host preparation use NumPy and SciPy. A compatibility probe also exercised cuBLAS, but the native Galerkin march uses project-specific projection kernels; cuBLAS should not be credited as the production projection backend.

Independent qualification reconstruction can use several CPU workers while a single owner captures GPU trajectories. Those elapsed times are excluded from method rankings. The intended latency experiment instead runs one GPU solve at a time with fixed affinities, an identical

Table 3: Recorded environment of the archived Windows experiments. Version labels describe the measured environment, not a recommendation to use the latest available release.

Component	Recorded value
CPU	AMD Ryzen 9 9955HX, 16 cores, 32 logical processors
GPU	NVIDIA GeForce RTX 5060 Laptop GPU; compute capability 12.0
Operating system / driver	Windows 11, build 26200; NVIDIA 596.13, WDDM
Scientific runtime	Python 3.13.5; NumPy 2.2.6; SciPy 1.16.3
Main CUDA components	CUDA 11.8 NVRTC and cuFFT; driver API version 13020
Precision and host threading	FP64 arithmetic; one BLAS thread in controlled comparisons
Main observation / nonlinear path	Warp probe reduction / original batched cuFFT

Table 4: Spatial configurations and eligible step-rate schedules. An arrow denotes the change at $t = 8$.

Configuration	Rank	K/q	Step rate
Mid RFM37	39	72/66	128 or 128 $\rightarrow$ 112
Mid Legendre35	39	72/66	128 or 128 $\rightarrow$ 112
High RFM41	40	80/76	152 or 152 $\rightarrow$ 136
High RFM35	38 or 39	80/76	152 or 152 $\rightarrow$ 136
High Legendre36 / 35	40 / 39	80/76	152 or 152 $\rightarrow$ 136
High Legendre34	38	80/76	152 only
FD176 dense / FD184 banded	—	$K = 80$	128 throughout

runtime policy, and sampled external-load checks. Low total CPU utilization during this experiment is expected: one controlled host core is only one of 32 logical processors. Concurrent throughput is a different objective and was not measured by this campaign.

8 Accuracy, trial-space size, and nonlinear workload

8.1 Qualified configurations and retained ranks

Table 4 lists the main configurations. The RFM and Legendre suffixes denote input basis counts, whereas the retained rank is obtained after enrichment, scaling, and truncation. The mean block retains rank 30 throughout. A finite-difference label n counts radial intervals, giving $n+1$ nodes and $n-1$ interior unknowns per angular mode. These interval counts are not directly comparable to a Galerkin retained rank.

There are 39 unique numerical identities, each evaluated under two explicit compilation-cache policies, for 78 GPU trajectories through $T = 72$. Every trajectory passed the registered physical, implementation, and wake criteria. Within each of the 39 cache-policy pairs, saved states, times, and probe arrays were bitwise identical. These are 39 numerical configurations reproduced under two policies, not 78 independent random-feature samples.

Across the full cohort, the maximum direct physical error is 0.477543%, and the maximum safeguard is 0.497730%. The latter leaves only approximately 0.002270 percentage points below the 0.5% limit. It is a measured acceptance margin rather than evidence of robustness to changes in Reynolds number, boundary conditions, or floating-point environment. The largest velocity error across full and near fields is 0.10497%; thus the tightest constraints arise from vorticity. This explains why velocity-only validation would have admitted a substantially weaker test.

The largest GPU-to-CPU field error is 3.5721×10^{-6} and the largest probe-series error is 3.8792×10^{-6}, both below 10^{-5}. These maxima correspond to the finite-difference control. For the final rank-expanded cases lacking a historical serial-observer control, the qualification still

Table 5: Worst errors by method and schedule, in percent. Each entry includes all registered seeds, both cache policies, the references, reconstruction grids, and checkpoints. Velocity and vorticity include full-domain and near-field checks.

Configuration	Schedule	Velocity	Vorticity	Guard
RFM37 / K72	Uniform	0.02943	0.40960	—
RFM37 / K72	Piecewise	0.04467	0.43168	0.47865
Leg35 / K72	Uniform	0.02676	0.44207	—
Leg35 / K72	Piecewise	0.04104	0.45665	0.49773
FD176 / K80	Uniform	0.10497	0.40876	—
RFM41 / K80	Uniform	0.01819	0.33189	—
RFM41 / K80	Piecewise	0.02595	0.34034	0.36867
Leg36 / K80	Uniform	0.01613	0.36205	—
Leg36 / K80	Piecewise	0.01972	0.36634	0.39405
Leg35 / K80	Uniform	0.01896	0.41520	—
Leg35 / K80	Piecewise	0.02235	0.41914	0.44722
FD184 / K80	Uniform	0.07900	0.30652	—
RFM35 / K80	Uniform	0.01935	0.42818	—
RFM35 / K80	Piecewise	0.02309	0.43245	0.46494
Leg34 / K80	Uniform	0.02301	0.47754	—

Table 6: High-resolution RFM35 by seed. Errors and safeguards are percentages, recomputed from the archived qualification scores; the two cache policies give the same arrays.

Seed	Rank	Constant-step E_ω	Piecewise E_ω	Piecewise safeguard
7331	38	0.427183	0.431520	0.463945
7687	38	0.427945	0.432246	0.464707
8011	38	0.428183	0.432451	0.464943
8353	39	0.378031	0.383875	0.414801
9041	38	0.427611	0.431910	0.464372

uses the registered CPU and reference comparisons; an unavailable earlier observer pairing is not silently treated as additional evidence.

8.2 Seed variation and the deterministic alternative

The five RFM seeds are fixed before the final qualification: 7331, 7687, 8011, 8353, and 9041. Table 6 exposes a confounding factor that is hidden if only the input label RFM35 is reported. Four seeds retain 38 radial functions and one retains 39. The smaller vorticity error for seed 8353 therefore occurs in a larger retained space. It cannot be attributed solely to a better draw of equal-dimensional features. Five fixed seeds also do not support a probabilistic guarantee over the feature distribution.

The rank-38 Legendre34 constant-step method also qualifies. Thus the evidence supports compact random-feature and deterministic polynomial spaces at this tolerance, but does not establish that randomness is necessary for compactness. A fair cost comparison must retain Legendre34 even though its piecewise variant fails, and must include Legendre35 and Legendre36 to separate equal-rank from accuracy-matched contrasts. High-resolution Legendre33 fails the $T = 72$ accuracy tests; a short-horizon pass is insufficient for promotion. The reduced mid-resolution RFM35 screen also does not replace the registered five-seed RFM37 cohort.

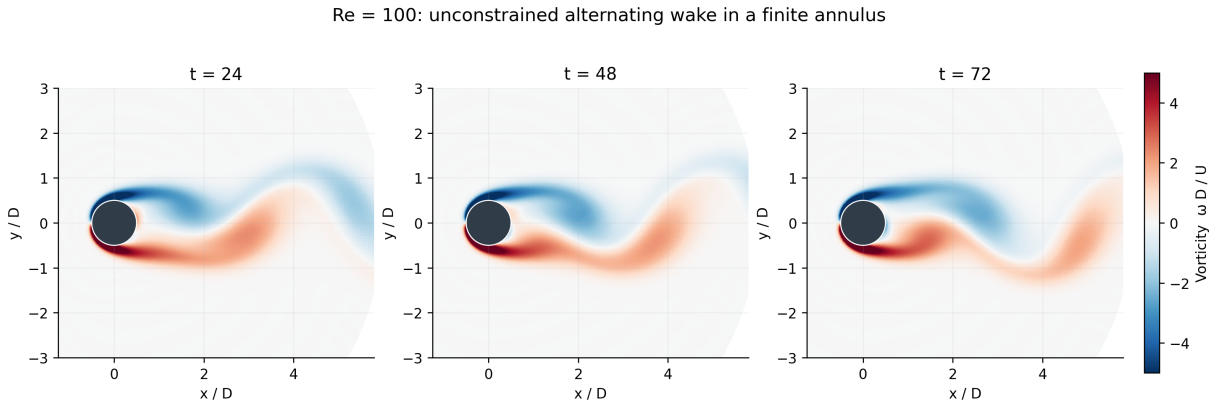


Figure 2: Vorticity at $t = 24$, 48, and 72 for one qualified RFM41 trajectory, without imposed reflection symmetry. The common color scale covers a local view of the annulus; the outer radius is 6.

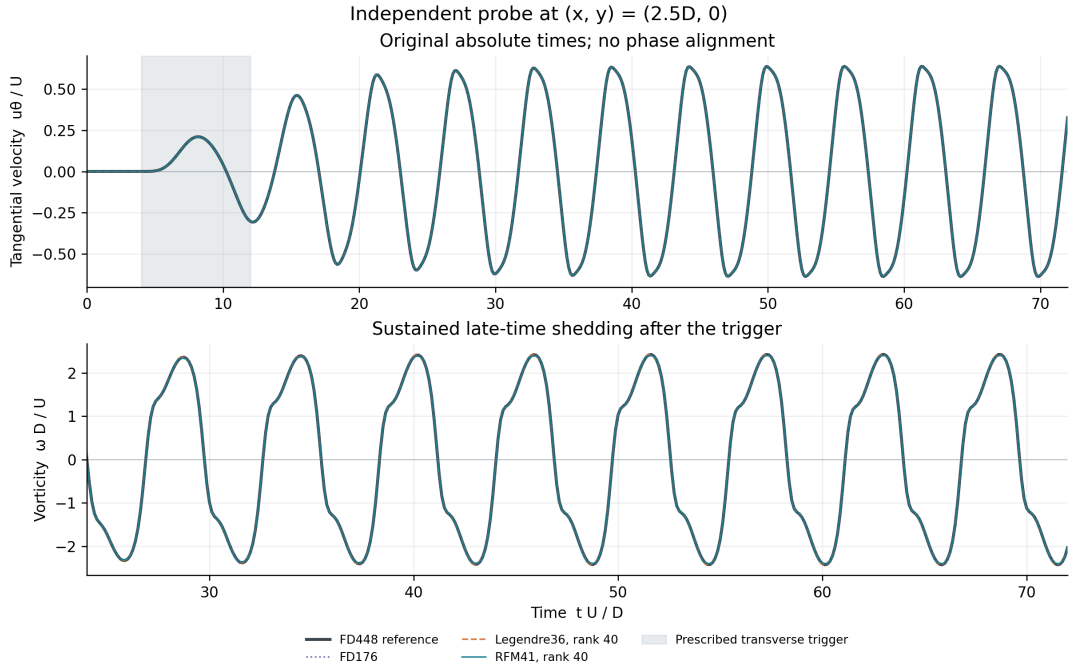


Figure 3: Unshifted probe histories at $x = 2.5D$, $y = 0$. Alternating shedding persists after the triggering pulse. RFM41 and FD176 are qualified GPU trajectories, Legendre36 is its corresponding CPU qualification, and FD448 is the reference.

8.3 Wake dynamics and the cost removed by scheduling

Figures 2 and 3 show the retained asymmetry and alternating shedding. The images display a local window, while scoring covers the prescribed full and near domains. The probe plot combines qualified trajectories and a fine reference for physical interpretation. Its curves are not from a single isolated timing batch and are not used to estimate errors by inspection.

The counts in Table 7 are read from runtime metadata rather than inferred from a nominal order label. Over the remaining 64 time units, reducing the step rate by 16 removes 1,024 steps; the recorded nonlinear counters differ by exactly the same amount. This saving applies to both RFM and Legendre. Reducing retained radial rank does not reduce q or the angular FFT length in these experiments, and therefore does not additionally remove FFTs from each nonlinear evaluation. The work count is an algorithmic saving, while its conversion into elapsed time depends on preparation, linear algebra, observation, and runtime costs.

Table 7: Recorded nonlinear evaluations for the two piecewise schedules. Both switch at $t = 8$ and end at $T = 72$.

Step-rate schedule	Constant step	Piecewise	Reduction
$128 \to 112$	9,215	8,191	1,024 (11.11%)
$152 \to 136$	10,943	9,919	1,024 (9.36%)

9 Additional ablations, error decomposition, and switch sensitivity

This section retains the 20 CPU trajectories added in revision 3: 16 ablation/control runs and four delayed-switch runs. Revision 4 adds diagnostics of saved probes and fields, without new PDE integration. The original 78 GPU qualification records remain frozen. The additional comparisons use the existing two references and physical checkpoints; passing their field tests does not automatically confer promotion under the full probe and GPU-implementation protocol. Concurrent CPU durations are excluded from speed rankings.

9.1 Additional ablation of frequency jitter and nonzero-mode layers

We fixed two configurations, $(m, K, q, \Delta t^{-1}) = (37, 72, 66, 128)$ and $(41, 80, 76, 152)$, before executing a two-by-two ablation: registered random paired-frequency jitter $\xi_j \sim U[0.2, 0.8]$ versus fixed midpoints $\xi_j = 0.5$, and six nonzero-mode analytic outer-layer features present versus absent. The random cells use the five registered seeds (7331, 7687, 8011, 8353, 9041). Each deterministic cell has one unique trajectory; unused seeds are not counted as replicates. All cells retain the same 24 mean-mode functions, six mean-mode layers, quadrature, and time step. Adapter checks confirm bitwise equality of the mean-mode discretization arrays and implicit operators across the four cells.

Revision 3 computed 16 CPU trajectories to $T = 72$: 14 ablation trajectories and two enriched-random CPU controls. Ten frozen enriched-random GPU trajectories were reused and independently rescored. Every trajectory was evaluated at the same 19 physical checkpoints on three grids against both fine FD references, using velocity and vorticity errors over the full domain and $r \leq 2.5$. All four errors must be at most 0.5%. Table 8 gives maxima over seeds, times, grids, and references. All 16 CPU runs completed without numerical failure; the 12 trajectories without nonzero-mode layers failed the accuracy gate. The two CPU controls differ from the archived GPU fields by less than 2.0×10^{-10} in the worst field metric.

Table 8: Additional CPU accuracy ablation. Errors are percentages. Pass counts exclude the two separate CPU controls. Random rows report the worst of five seeds; midpoint rows contain one deterministic configuration. The two enriched-random CPU controls have relative field-metric differences from their archived GPU controls below 2.0×10^{-10}; this bound does not cover the other ablation arms.

Configuration	Rank	e_u	e_ω	$e_{u,\mathrm{near}}$	$e_{\omega,\mathrm{near}}$	Pass
Mid, Random+layer	39	0.0294	0.4096	0.0251	0.2100	5/5
Mid, Midpoint+layer	39	0.0293	0.4092	0.0249	0.2097	1/1
Mid, Random/no layer	34	19.4054	73.8980	26.4854	48.6306	0/5
Mid, Midpoint/no layer	34	19.3644	73.7836	26.4397	48.5825	0/1
High, Random+layer	40	0.0182	0.3319	0.0148	0.1799	5/5
High, Midpoint+layer	40	0.0181	0.3312	0.0147	0.1794	1/1
High, Random/no layer	35	18.7540	72.0115	25.7262	47.7640	0/5
High, Midpoint/no layer	35	18.7060	71.8705	25.6678	47.6945	0/1

Both midpoint-plus-layer configurations pass, with errors comparable to the random enriched

cells. These data therefore do not establish randomness as necessary for the observed accuracy. With raw m and the SVD cutoff fixed, removing the nonzero-mode layers reduces retained rank from 39/40 to 34/35 and produces large velocity and vorticity errors; the mean rank remains 30. The compact configurations depend strongly on analytic outer-layer enrichment, but this design does not separate enrichment geometry from the increased effective dimension. It is not an equal-rank enrichment ablation, and only one deterministic frequency design was tested. All four error maxima for both deterministic no-layer configurations occur at $T = 72$. Their maximum vorticity errors for $t \leq 16$ are 2.820% and 2.592%, growing to 73.784% and 71.871% at later times. The large errors therefore are not solely a consequence of a near-zero startup reference field; the mechanism of error amplification is not established by these data. A subsequent construction-only check increased raw m to 41, 45, 49, 53, 57, and 61 while retaining bandwidth 62, quadrature, and the SVD cutoff. Both resolutions produced ranks 35, 35, 36, 36, 36, and 36, respectively, never recovering rank 39 or 40. We therefore did not add a purported equal-rank $T = 72$ control or tune the bandwidth/cutoff after failure. Concurrent CPU elapsed times are diagnostic only: they provide neither isolated timing nor new GPU speedup evidence, and do not extend the original 78 GPU qualification trajectories.

What the enriched-space comparison does and does not identify. The Legendre35 and Legendre36 controls also contain the same six analytic layers in Eq. (3.9). The present experiment therefore does not show that an unenriched Legendre space succeeds where an unenriched trigonometric space fails. Both families use the same prescribed radial quadrature at each resolution; a Legendre trial basis does not automatically replace it with a family-specific Gauss–Lobatto grid. Endpoint clustering of polynomial interpolation nodes is consequently not a measured explanatory variable in this comparison. An unenriched, rank-matched polynomial control would require a separate experiment.

The layer functions decay as $e^{-\gamma(L-s)}$ and are concentrated near the *outer* boundary $s = L$, not the cylinder wall $s = 0$. Moreover, the homogeneous far-field operator uses $\chi = 16z^2(1-z)$, with the correction in Eq. (3.8). Its no-slip conditions do not force the inner-wall curvature to vanish: with primes denoting z derivatives, $(\mathcal{B}q)''(0) = 32q(0) - [(\chi q)'']_{z=1}$, which is generally nonzero. Thus removing these layers cannot by itself establish a chain from lost inner-wall curvature to a changed separation angle. The large final errors demonstrate sensitivity to the trial space and retained dimension together. Likewise, a midpoint design matching or slightly improving the worst error of five sampled seeds is not a theorem that randomization never helps.

Saved-probe evidence for accumulated phase drift. We analyse the existing 577 observations at spacing 0.125, without integrating a new trajectory or shifting the physical scoring times. The primary window is $24 \leq t \leq 72$. After subtracting a least-squares linear trend, upward zero crossings are linearly interpolated and their times are fitted to an affine function of crossing index; the inverse slope estimates frequency. Both velocity and vorticity at $x = 1.5, 2.5, 3.5$, $y = 0$ are retained, with windows starting at 36 and 48, mean-only detrending, and both finer FD references as sensitivity checks. These finite-window estimates are descriptive, not confidence intervals or an unbounded-cylinder Strouhal validation.

Across all five random seeds and the midpoint controls, the primary-window no-layer frequency biases range from -4.27% to -3.35% over the six probe signals; including both references, windows and detrending variants expands this range to -5.20% to -1.88%. The observed delays in Figure 4 support accumulated phase drift as one component of the long-time mismatch. For the two no-layer midpoint rows, $2\pi(f - f_{\mathrm{ref}})48$ equals approximately -2.22 and -2.10 radians, a rate-based estimate rather than a measured full-field correction. The data do not support assigning all 70%-level field error to phase. The removed enrichments are localized at the *outer* boundary, retained rank changes simultaneously, and no inner-wall separation angle was measured. An inner-wall cause is therefore unestablished.

Table 9: Stored midpoint-frequency trajectories: vorticity at $(2.5, 0)$ over $[24, 72]$, using the time-fine FD reference. Each fit uses eight upward crossings (seven complete intervals). Reference frequency is 0.17534308. Bias is relative frequency error in percent; delay is the candidate-minus-reference time at the matched reference crossing $t = 66.797754$.

Resolution	Nonzero layers	Rank	Frequency	Bias (%)	Delay
Mid	Present	39	0.17534574	+0.00152	−0.00101
Mid	Absent	34	0.16799056	−4.19322	+1.75259
High	Present	40	0.17534385	+0.00044	−0.00020
High	Absent	35	0.16837253	−3.97538	+1.67221

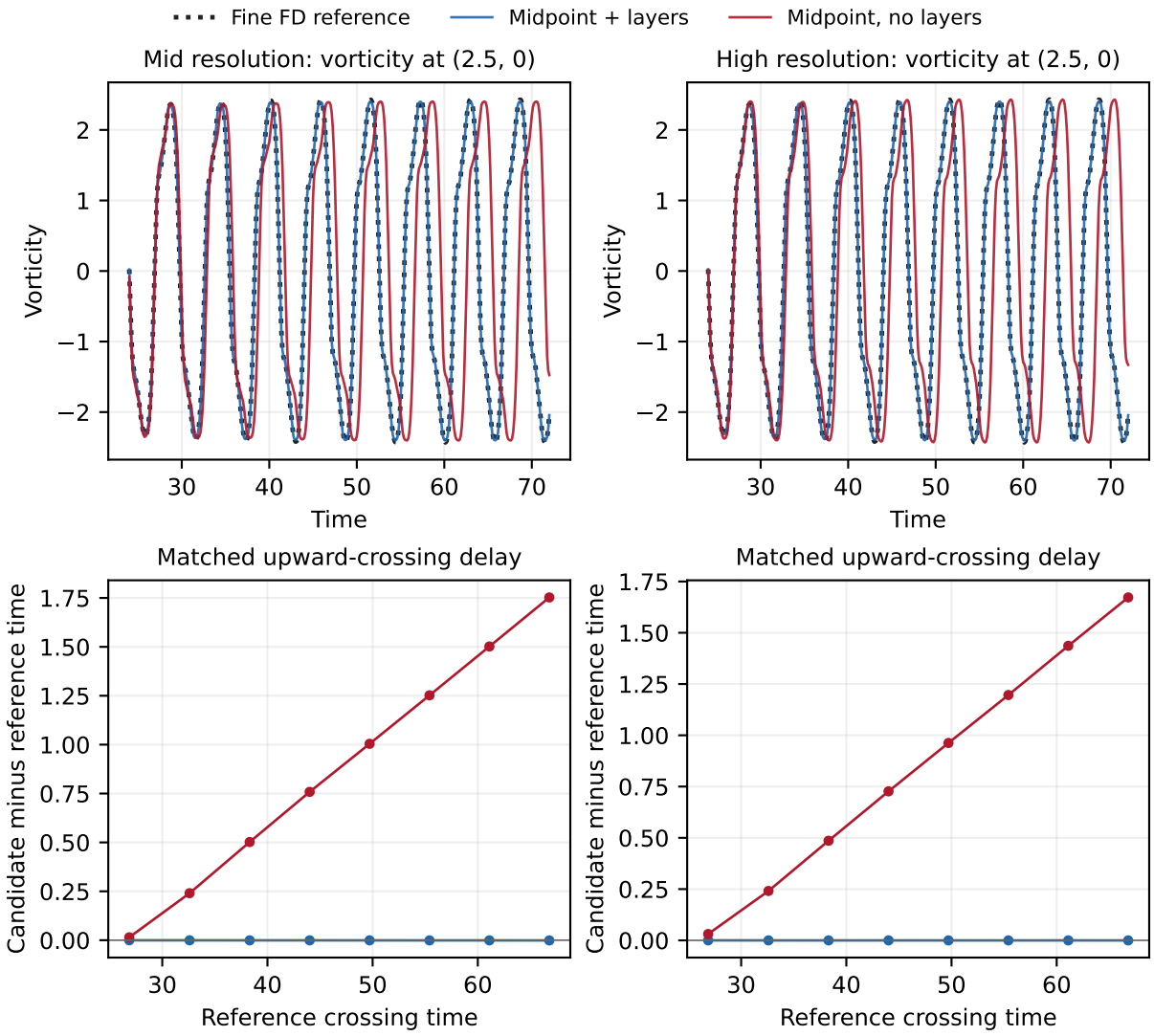


Figure 4: Original-time vorticity signals and accumulated crossing delays for the midpoint-frequency ablation. The first candidate crossing is paired with the nearest initial reference crossing, after which crossing indices remain fixed. The lower panels use linearly detrended signals for event detection only; no trajectory or field-error score is time shifted.

9.2 Matched-time radial and angular error diagnostics

The maximum errors in the qualification tables need not occur at the same time, grid, or reference. We therefore reanalyse the hash-bound CPU controls corresponding to the qualified high-resolution RFM41 (seed 7331), Legendre36, and FD184 trajectories. The first two use the original $t_s = 8$ schedule and retain the same nonzero-mode rank 40. The uniform RFM41 and Legendre36 controls are also included in the machine-readable results. Both existing FD448/$K = 144$ references are reconstructed: `angular_fine` uses rate 384 and `time_fine` rate 512. No reference or candidate is integrated for this analysis. All 19 physical times are retained; the figures show the common times $8, 24, 48, 72$ against `time_fine`. There is no phase alignment.

Let e_f be the velocity-vector or scalar-vorticity error and let $\langle\cdot\rangle_\theta$ denote the angular mean. For the existing layer144 radial nodes and physical-area weights w_i, define

$$R_f^2 = \frac{\sum_i w_i \langle |f_{\mathrm{ref}}(r_i,\cdot)|^2\rangle_\theta}{\sum_i w_i}, \qquad P_f(r_i) = \frac{\langle |e_f(r_i,\cdot)|^2\rangle_\theta^{1/2}}{R_f}. \tag{9.1}$$

This normalization avoids dividing by a locally vanishing field. For real fields represented by nonnegative Fourier modes, the angular error contribution is

$$Q_{f,k} = \left(\frac{c_k \sum_i w_i |\widehat{e}_{f,k}(r_i)|^2}{\sum_i w_i \langle |f_{\mathrm{ref}}(r_i,\cdot)|^2\rangle_\theta}\right)^{1/2}, \qquad c_0 = 1, \quad c_k = 2\ (k>0). \tag{9.2}$$

For velocity, squared component magnitudes are summed in the polar representation (u_r, u_θ). The integrated vector-error energy is invariant under the pointwise orthogonal change to Cartesian components, but its allocation among angular modes depends on that representation; the displayed velocity spectrum is therefore a polar-component spectrum. Hence the weighted mean of P_f^2 and the sum of $Q_{f,k}^2$ both give the squared relative field error. The exact angular Parseval calculation agrees with the original layer144 physical-grid score to 2.13×10^{-17} in 380 checks. Modes above the candidate cutoff $K = 80$ contain the common unresolved reference tail, not additional candidate degrees of freedom.

The original maxima are not a simultaneous ranking. For this RFM seed, the full-domain velocity and vorticity maxima occur at $t = 72$ and 60; the corresponding Legendre maxima occur at 60 and 54. In contrast, at the same $t = 48$ and on layer144, RFM41, Legendre36, and FD184 have velocity errors 0.01886%, 0.01842%, and 0.02053%, but vorticity errors 0.29438%, 0.35509%, and 0.11317%. Thus even a simultaneous velocity ranking does not determine the vorticity ranking. Curl depends on first derivatives of velocity; an L^2 velocity comparison alone does not control their error. This observation is consistent with the measured spectra: at $t = 72$, modes $k \le 8$ contain 84.24% of FD184's squared velocity-error norm, whereas modes $25 \le k \le 80$ contain 47.72% of RFM41's squared vorticity-error norm, compared with 5.45% for FD184.

The radial localization also requires qualification. At $t = 72$, 61.14% of RFM41's squared vorticity-error norm lies at $r \ge 4$, but only 1.37% lies at $r \ge 5.9$. The observed error is therefore not confined to the narrow outermost layer. These diagnostics localize error in the full evolved solutions. They do not identify a causal benefit of the layer basis, or separate radial approximation, time discretization, boundary treatment, and reference uncertainty. The accompanying controlled ablation addresses basis attribution; the present figures alone do not.

9.3 A CPU sensitivity experiment at $t_s = 12$

Revision 3 added four actual CPU trajectories through $T = 72$: RFM37 and Legendre35 at rates $128 \to 112$, and RFM41 and Legendre36 at $152 \to 136$, switching at $t_s = 12$ instead of 8. RFM uses seed 7331; the deterministic controls retain their original configurations. The generic CPU phase clock already accepts an explicit switch time. Its default observation list contains the old $t_s = 8$ bridge instants, so this driver supplies the 19 common physical times explicitly. The

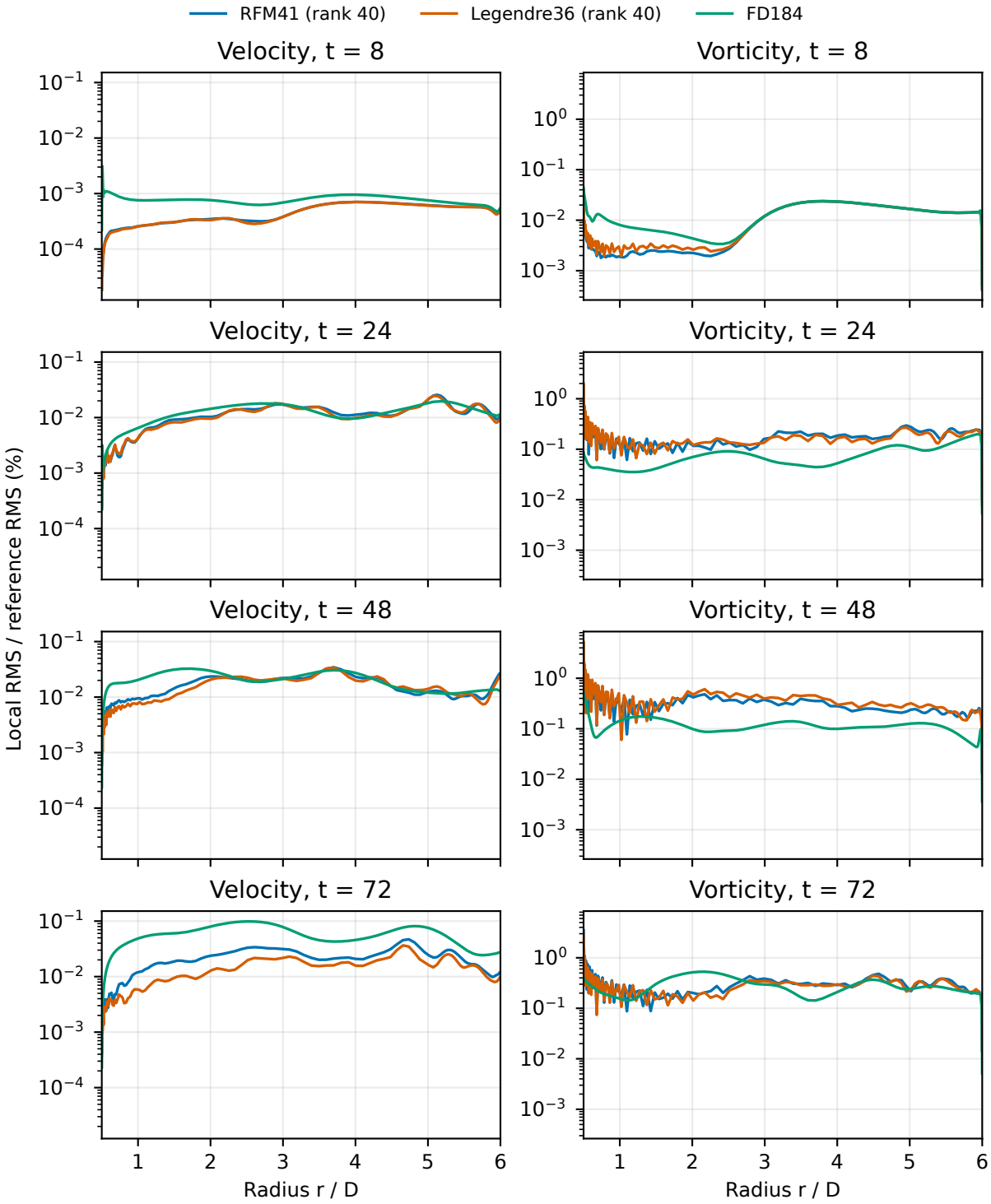


Figure 5: Matched-time radial error profiles $100P_f$ against the existing time-fine reference. RFM41 uses seed 7331. Galerkin curves use $t_s = 8$; FD184 retains its qualified schedule. Curves share the same reference and times within each panel. The plotted quantity is a local angular RMS normalized by a global reference RMS, rather than a pointwise relative error.

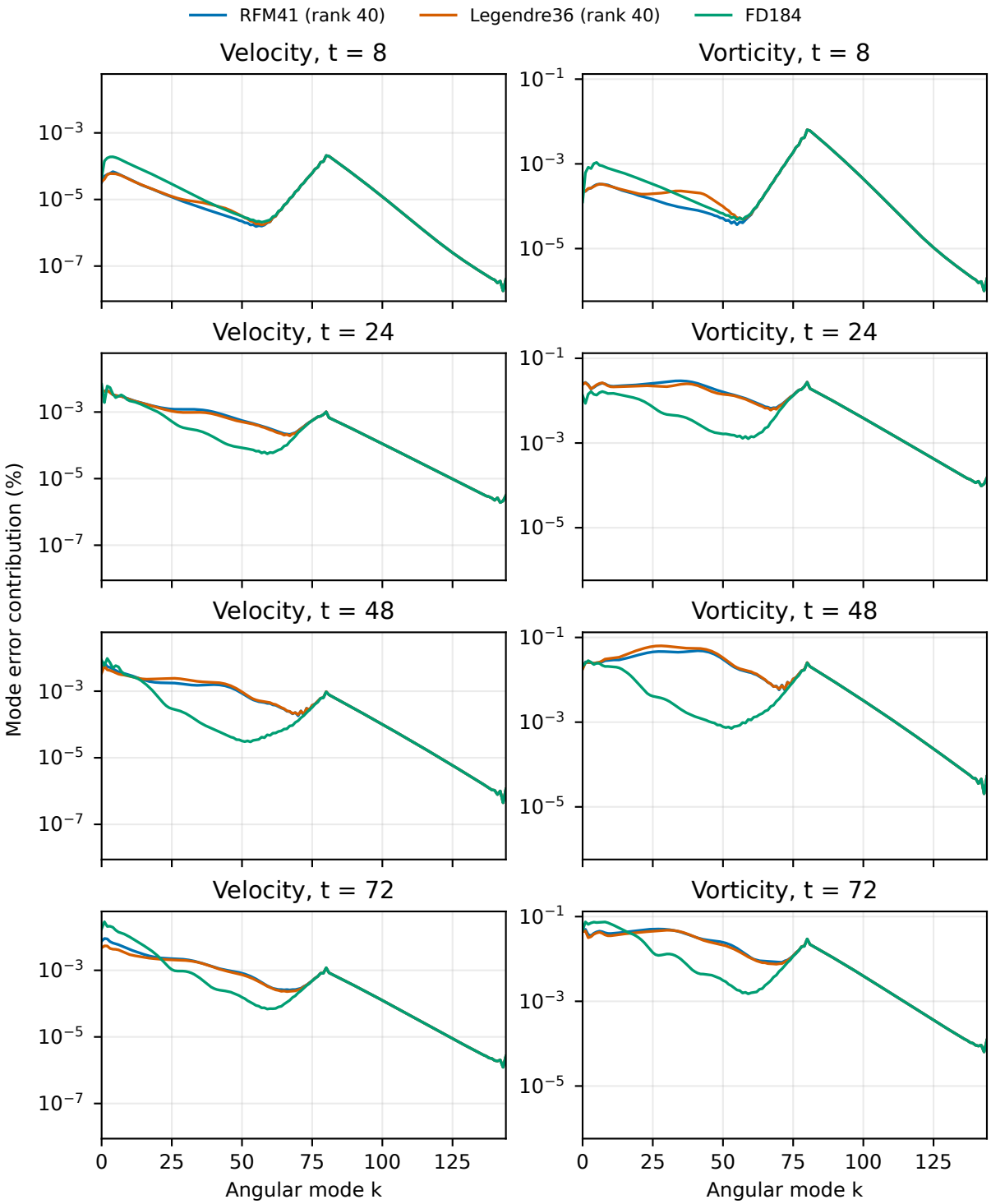


Figure 6: Angular error contributions $100Q_{f,k}$ for the same trajectories and times as Figure 5. Contributions combine in quadrature; they are not independently normalized modewise errors. The common tail above mode 80 reflects the finer reference spectrum.

underlying solver and archived data are unchanged. Four workers each use one BLAS thread; their elapsed times are not isolated performance evidence. Existing uniform and $t_s = 8$ CPU controls are reused, rather than rerun or counted as new experiments.

All four new trajectories match the corresponding uniform CPU states exactly at the saved times $t \leq 12$. Both full-domain and near-field velocity and vorticity pass the 0.5% threshold on all three grids against both frozen references; the per-cell triangle safeguard relative to the matching uniform control also passes, and alternating probe signs persist. Table 10 reports full-domain field maxima separately from the worst safeguard across all four metrics. These are one-seed sensitivity results, not a new five-seed GPU qualification.

Configuration	Schedule	E_u (%)	E_ω (%)	Guard (%)	N_{nl}
RFM37	Uniform	0.029428	0.409349	0.409349	9215
	$t_s = 8$	0.044672	0.431680	0.478647	8191
	$t_s = 12$	0.043821	0.429910	0.474465	8255
Legendre35	Uniform	0.026763	0.442065	0.442065	9215
	$t_s = 8$	0.041043	0.456649	0.497730	8191
	$t_s = 12$	0.040211	0.455435	0.493773	8255
RFM41	Uniform	0.018192	0.331380	0.331380	10943
	$t_s = 8$	0.025948	0.339890	0.368159	9919
	$t_s = 12$	0.025496	0.339258	0.365822	9983
Legendre36	Uniform	0.016132	0.362049	0.362049	10943
	$t_s = 8$	0.019717	0.366338	0.394046	9919
	$t_s = 12$	0.019410	0.365962	0.391767	9983

Table 10: CPU switch-time sensitivity at the frozen spatial configurations. Errors maximize over two references, three grids, and 19 common times; each metric has its own argmax. The guard also includes the near-field metrics. RFM rows use seed 7331 and therefore need not equal the earlier five-seed maxima. N_{nl} counts actual nonlinear evaluations; the number of marching FFT batches is $2N_{\mathrm{nl}}$. No GPU trajectory is available for $t_s = 12$, so no CPU–GPU bound is assigned to those rows.

Moving the switch from 8 to 12 adds 64 nonlinear evaluations and 128 FFT batches in every configuration. It removes 960 evaluations relative to the uniform schedules: 10.418% at the middle scale and 8.773% at the high scale, versus 11.112% and 9.358% for switching at 8. The measured error decrease is modest: for RFM37 the velocity maximum decreases by 0.000851 percentage points and the vorticity maximum by 0.001771 percentage points. The closest new safeguard is 0.493773% for Legendre35. The later switch therefore trades a small increase in work for a small accuracy margin on this case; no isolated speed ranking follows.

The triggering transverse pulse terminates at $t = 12$. Its $\sin^4$ profile is C^3 at that junction, with a possible fourth-derivative jump. The local Taylor consistency calculation for the bridge must respect the available one-sided regularity there. This experiment is a switch-time sensitivity test, not evidence that a smooth-transient order reduction has been removed. No timestep-refinement series is supplied to estimate an observed order. Separately, the archived 10^{-5} implementation gate compares CPU/GPU or CPU operator implementations of the *same* discrete schedule. It is not a tolerance for changing from uniform to mixed timesteps; that change is assessed by the physical errors and the 0.5% safeguard. In the quoted middle-scale RFM comparison, the velocity maxima 0.029428% and 0.044672% differ by 0.015244 percentage points (about 51.8% relatively), and both happen to occur at $t = 72$. This relative increase alone establishes neither a transient-splitting mechanism nor order reduction.

A norm reduction is not a causal error budget. For the middle RFM velocity maxima, postponing the switch recovers approximately 5.584% of the increase relative to the uniform schedule. This is the arithmetic ratio $(E_8 - E_{12})/(E_8 - E_U)$; it does not isolate a bridge-local

error. The intervention simultaneously changes four time units of fine versus coarse integration, the bridge time and state, and the subsequent error history. Moreover, $t = 8$ is inside the analytic pulse, not a nonsmooth forcing junction.

The saved fields make the nonadditivity explicit. Let $e_8 = u_8 - u_{\mathrm{ref}}$, $e_{12} = u_{12} - u_{\mathrm{ref}}$ and $\delta = e_8 - e_{12}$, with a common physical-area inner product divided by $\|u_{\mathrm{ref}}\|^2$. Then

$$\|e_8\|^2_{\mathrm{ref}} = \|e_{12}\|^2_{\mathrm{ref}} + 2\langle e_{12}, \delta\rangle_{\mathrm{ref}} + \|\delta\|^2_{\mathrm{ref}}. \tag{9.3}$$

At $t = 72$ on layer144 against the time-fine reference, for RFM37 seed7331 these four terms are $1.9956194141 \times 10^{-7}$, $1.9202943907 \times 10^{-7}$, $7.4475738974 \times 10^{-9}$ and $8.4928450335 \times 10^{-11}$, respectively. They are squared relative errors, without percent scaling. The unrounded identity residual is 2.04×10^{-22}; the cross term accounts for about 98.87% of the squared-norm difference. This deterministic identity illustrates why neither the 5.6% recovery nor its 94.4% complement is an independent physical error component. A causal bridge/coarse-step split would need separately controlled interventions and a timestep-refinement study.

10 Operator performance and the cost of a complete solve

10.1 FFT fusion: a local improvement with first-use overhead

Table 11 measures repeated evaluations on a fixed state with hot input arrays. It excludes model construction, compilation, module loading, FFT planning, and physical trajectory advancement. The fused implementation improves all four measured nonlinear operators. Its advantage is largest for the smaller Galerkin batch; the table does not establish a general advantage over cuFFT on arbitrary transforms or hardware. The comparison is between complete nonlinear pipelines with the same output definition, rather than a single custom FFT and an unrelated library call.

The eight subsequent $T = 9$ trajectories all passed the reference and implementation tests. Relative to the same warp-observer cuFFT path, the maximum field difference was approximately 1.78×10^{-9}; the maximum difference from the independent CPU trajectory was approximately 2.26×10^{-7}. Marching was 9%–17% faster. Nevertheless the first complete solve was 23%–69% slower. Compilation and module loading increased by approximately 0.25–0.62 s, while marching saved only 0.017–0.030 s. Figure 7 makes this reversal visible without adding overlapping timing categories.

Caching changes this tradeoff but did not remove the admission requirement. In the later strict-hit diagnostics, all four complete solver clocks improved, whereas only two of the four fresh-process comparisons improved. The candidate therefore failed its registered promotion rule and was excluded from the final $T = 72$ cohort. This is an engineering selection failure, not a failure of its short-trajectory numerical tests. Extrapolating the local ratio to a long trajectory would neglect both overhead and changes in the distribution of work.

Table 11: Fixed-state CUDA graph event measurements in microseconds per nonlinear evaluation. Each configuration uses nine samples of 64 evaluations. Ratios refer only to this operator measurement.

Configuration	Original cuFFT	Fused	Local ratio
Galerkin $N = 225$, $q = 66$	42.12	22.64	1.86
Galerkin $N = 243$, $q = 76$	40.32	25.48	1.58
FD176, $N = 243$	85.13	61.00	1.40
FD184, $N = 243$	90.35	69.38	1.30

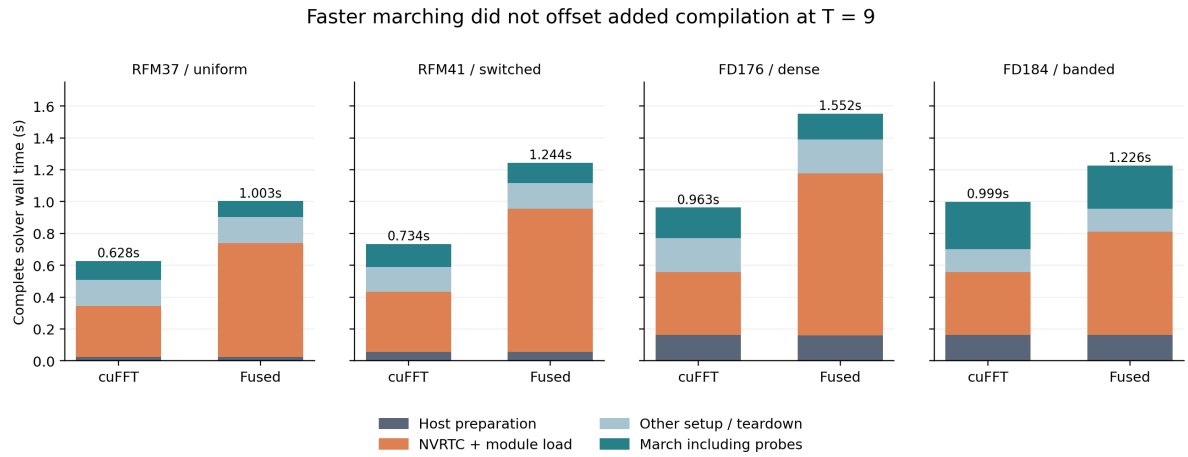


Figure 7: Cost decomposition of the sequential $T = 9$ fusion diagnostics. Compilation and loading are separated from the remaining runtime initialization. These diagnostic measurements identify first-use costs; they do not provide the missing final $T = 72$ ranking.

10.2 Observation kernels and common implementation benefits

Warp-based reduction improves a different part of the pipeline. Fixed-state probe-only nodes decrease from approximately 117–147 μs to 6–7 μs for the tested Galerkin paths, from 727.5 to 26.2 μs for FD176, and from 760.3 to 28.3 μs for FD184. The resulting local ratios are approximately 19–27. Five full $T = 72$ checks preserve the saved states and times bitwise; the changed summation order gives a maximum relative probe difference of 9.26×10^{-14}. The operator does not alter the evolution equations.

Because the original FD observation node is more expensive, giving this optimization only to the proposed Galerkin solver would distort the comparison. The final cohort uses the warp observer for all relevant baselines. Multiplying hot-kernel savings by the observation count can suggest where time is spent, but it is not a substitute for the complete paired timing of the modified solver. The local ratio must not be quoted as a 19–27-fold solver speedup.

10.3 Cache policy, startup, and amortization

The explicit cache stores PTX compiled from project kernels; it does not store a precomputed solution, spatial matrices, CUDA context, or cuFFT plan. The key binds source, compiler options, target architecture, entry points, compiler identity, and dependency hashes. Under `strict_hit`, a missing or corrupt artifact is an error, with no silent compilation fallback. Under `empty_first`, each run begins with a new directory and pays for compilation, atomic writing, readback verification, and loading. Driver and operating-system caches remain in their natural state. Thus "first use" is an explicit policy, not a claim that the whole machine has been reset to a cold state.

Three stock PTX keys occupy 147,782 bytes. Three fresh-process installations take 2.7292, 2.6392, and 2.6487 s, with a median of 2.6487 s. Reporting only the roughly 0.93 s NVRTC stage would omit most of this installation path. If C is the complete installation cost, H a subsequent hit cost, and F a per-solve first-use cost under otherwise matched conditions, the amortized comparison is

$$T_{\text{cache}}(n) = C + nH, \qquad T_{\text{first}}(n) = nF. \tag{10.1}$$

A crossover requires $F > H$ and $n > C/(F - H)$. The final campaign does not provide an accepted paired estimate of $F - H$, so no measured break-even solve count is reported. The installation covers three keys; the minimum installation cost for a user needing only one key was not measured.

Table 12: Local phase-preparation measurements in milliseconds. These use the prototype's original anchor, not the later geometric-anchor rule.

Configuration	Original	Triangular solves	Common phase basis
Mid RFM	34.27	47.41	51.96
Mid Legendre	34.81	44.72	47.73
High RFM	38.75	56.39	61.39
High Legendre	45.71	62.00	66.22

An earlier 50-run high-resolution campaign passed its sampled isolation check and measured median paired fresh-process speedups of 1.293 and 1.301 after import simplification for constant and piecewise RFM, while solver-time ratios stayed near unity. This identifies a startup improvement for that historical configuration. It does not replace a comparison with the stronger Legendre baseline, common warp observer, or explicit cache policies introduced later. Pooling these rows would mix different experimental treatments.

10.4 A conditional crossover for a longer time schedule

The historical single-bridge diagnostic records about $\Delta M_0 = 0.119$ s of marching savings and $\Delta P = 0.206$–0.240 s of additional preparation. These measurements come from a window marked `diagnostic_only`, with external activity, and precede the final configuration set. They cannot supply a measured final speed ratio. They can, however, make the assumptions of a crossover estimate explicit.

Let N count steps of the piecewise candidate, including the unchanged first $N_s = 1024$ steps. Its $T = 72$ trajectory has $N_0 = 8192$ steps. If the extra preparation stays constant and the saved marching time grows linearly with only the later work, the modeled candidate-minus-control time is

$$\Delta T(N) = \Delta P - \frac{N - N_s}{N_0 - N_s}\Delta M_0, \qquad N_* = N_s + (N_0 - N_s)\frac{\Delta P}{\Delta M_0}. \tag{10.2}$$

Substitution gives $N_* \simeq 13{,}432$–$15{,}480$ candidate steps. A coarser model that scales the savings with all steps instead gives $N_* = N_0 \Delta P / \Delta M_0 \simeq 14{,}181$–$16{,}522$. Their difference reflects the treatment of startup, not additional measurements. Thus a scale of roughly 1.5×10^4 steps is a conditional planning estimate, not an observed threshold. Neither model proves accuracy beyond $T = 72$, constant device efficiency, or superiority to a different qualified method. Multiple short solves amortize preparation only when the relevant prepared objects are actually shared; a PTX cache alone does not share spatial matrices or FFT plans. A new longer-horizon qualification and isolated experiment are required to validate a crossover.

11 Rejected candidates and the limits of local theory

11.1 Common phase coordinates and dense output layout

The common phase-basis prototype improves local linear-algebra diagnostics but, with the existing dense device interface, increases three-phase preparation by about 37%–58%. Table 12 reports the measured values. Device storage remains 11.69–16.30 MB because dense physical-coordinate operators are still generated. The prototype pays for eigendecomposition and back-transformation of wide right-hand-side blocks instead of remaining in diagonal coordinates throughout the computation.

The analytical anchor in Eq. (5.4) minimizes a condition-number upper bound. It was not timed as a new production candidate. Better transformed conditioning, fewer factorizations, less stored data, and shorter complete runtime are distinct propositions. A future implementation

that keeps state in common coordinates would need corresponding reconstruction, lifting, projection, and observation operators, followed by a new floating-point and trajectory qualification. The existing prototype does not test that architecture.

11.2 Energy structure does not guarantee coarse-step accuracy

The Lamb–BDF2 candidate freezes an extrapolated vorticity field and solves the resulting velocity system iteratively. Appendix B derives the discrete skew-symmetry and fixed-step energy identity, including boundary and algebraic-residual work. These properties concern the stated discrete update and norms. They do not eliminate the error introduced by extrapolating vorticity or establish accuracy at an arbitrarily large step.

At $T = 16$ and step rate 128, the worst vorticity errors are 0.364950% for RFM37 and 0.364784% for Legendre37, satisfying the 0.5% field criterion. At rate 64 they rise to approximately 161.024% and 161.664%. The first saved failing state occurs at $t = 4$, before the transverse pulse begins. At the recorded checkpoints the original-equation residuals remain at most 7.590×10^{-10}, and work-corrected energy defects remain at most 2.25×10^{-15}. A tightly solved linear system and a closed energy identity can therefore coexist with an inaccurate flow trajectory.

The qualifying rate-128 candidate requires approximately 5.03 times as many one-dimensional FFTs as the explicit control in the same space. This count includes products needed by the iteration and residual checks; halving nominal time steps alone would hide that work. The rate-64 variants also require more transforms and fail accuracy. The branch was consequently rejected before a $T = 72$ GPU comparison.

11.3 Skipping nonlinear evaluations and changing graph schedules

Evaluating the nonlinear term on alternate steps gives an appealing nominal reduction. Scalar tests of the corresponding history extrapolation show observed convergence orders of 1.998–2.016. However, corrected Navier–Stokes pilots that activate this strategy after $t = 8$ overflow near $t = 9.508$ and 9.559, while their ordinary controls complete. The failed trajectories do not supply valid accuracy or timing results. This experiment illustrates why local consistency and scalar stability probes are screening tools rather than guarantees for the driven, mode-coupled discretization.

Two implementation experiments were numerically successful but did not meet performance-selection criteria. Alternating the two existing spectral-history buffers preserves eight short paired trajectories bitwise but gives inconsistent marching changes of -0.030 and $+0.018$ ms in the graph-enabled pair summaries. Changing graph chunk length from 24 to 48, 64, or 128 steps preserves 36 full trajectories bitwise. Although 48 steps halves the scheduled graph launches, no alternative simultaneously meets the registered marching, solver, and fresh-process improvement criteria. These diagnostics lack full external-load isolation and do not establish small stable speed differences. The 24-step graph schedule is retained.

12 Disposition of the final timing experiment

The final design registers 160 slots across five paired blocks, two cache policies, and the qualified methods. RFM seeds are assigned by block; deterministic methods remain fixed. Both the complete solver clock and the fresh-process clock are needed for a practical result. The latter includes the declared supervisor and process-start costs. Every window must finish its prescribed sequence, satisfy numerical checks, and pass sampled isolation; failed slots are not replaced selectively with cleaner runs.

In the last attempted slot, the solver writes its numerical output and exits with a successful exact-process status, but a subsequent permission error prevents publication of the terminal

Table 13: Final timing attempts and admissible interpretation. None provides an accepted final paired speed ranking.

Attempt	Recorded outcome	Consequence
Original design	Two slots complete; the third has no confirmed timely terminal state	Incomplete paired blocks
First recovery	External activity in preflight; zero slots launched	Window rejected before timing
Final recovery	12 attempted; 11 raw successes; terminal publication fails for the twelfth	Completion requirement fails
Independent load check	External CPU activity exceeds the registered budget during the final active window	Isolation fails independently
Numerical readback	All 12 saved trajectories match their qualification arrays; 148 slots remain unattempted	Numerical evidence retained; no ranking

state. The available record does not identify the original lock holder. A later CPU-only diagnostic reproduces a file-publication failure under a controlled reader configuration, without proving that it caused the earlier event. Separately, the load monitor observes external CPU activity above 0.2 CPU seconds per ten-second budget interval. Correcting terminal publication alone would therefore not make the recorded batch admissible.

The isolation mechanism uses sampled process identities and preflight GPU observations, not exclusive ownership enforced by the operating system. Very short processes can escape between samples, and coverage of unrelated applications is limited. Those limitations apply even to historical windows that passed the sampled check. For the final window, the recorded violations are sufficient to reject the timing regardless of the additional unobserved activity that might have existed.

No final speedup, confidence interval, or method ranking is constructed from the 11 successful raw records, from qualification durations, or by joining the two earlier slots to a later recovery. The missing high-resolution windows also prevent a claim of stable behavior across resolutions. The result is an unresolved performance ordering, rather than a measured win or loss of RFM against the strongest classical baseline.

13 Discussion and conclusions

13.1 What changes on a nonseparable geometry

The annulus supplies more than a convenient test shape. Its metric and boundary conditions are independent of angle, so the implicit operator separates into angular blocks; the nonlinear products still couple modes. The retained radial rank d, number K of positive modes, quadrature size q, and padded angular size N lead to dense setup costs of order Kd^3 for independent modal factorizations. A step contains order Kd^2 modal solves, qdK reconstruction and projection work, and $qN \log N$ transform work. These counts suppress fixed channel counts and the separate mean block. They describe the present implementation, not a complexity theorem for RFM in general. Differential normalization of an n-column raw feature matrix additionally costs order $qn^2 + n^3$ in the dense regime $q \geq n$.

On an angle-dependent mapped boundary, the metric coefficients and boundary lifting generally depend on angle. Multiplication by these coefficients couples Fourier modes even in the implicit linear operator. The scalar mode equations and their separate common-phase bases then cease to be an exact factorization of the full problem. A naive dense solve with $D = O(Kd)$

coupled degrees of freedom would require $O(D^2)$ storage and $O(D^3)$ factorization. This is an avoidable dense-design cost, not a necessary lower bound: sparse local spaces, matrix-free metric products, domain decomposition, and iterative solvers are alternatives. They require a new operator design and qualification; the present GPU measurements do not test them.

The role of conditioning can be stated precisely without claiming that the annular transform removes it. Let B be an SPD reference implicit operator and A a mapped-domain SPD operator on the same discrete space. If the geometry and boundary construction yield constants $0 < \alpha \le \beta$ such that

$$\alpha x^\top B x \le x^\top A x \le \beta x^\top B x \quad \text{for all } x, \tag{13.1}$$

then the Rayleigh quotient gives

$$\sigma(B^{-1/2} A B^{-1/2}) \subset [\alpha, \beta], \qquad \kappa_2(B^{-1/2} A B^{-1/2}) \le \beta/\alpha. \tag{13.2}$$

This is a conditional preconditioning bound. We do not establish mesh-independent α, β for an arbitrary body. Distorted mappings, narrow gaps, and nonuniform local resolution can degrade them. Boundary-data approximation introduces a separate consistency error that is not controlled by this spectral bound; it need not alter the implicit matrix. Likewise, the identity $K_1 = \nu I$ on the retained annular feature space is an exact-arithmetic normalization at the chosen quadrature; it does not bound every modal mass matrix, discarded approximation direction, or floating-point reconstruction error.

Local random features with partitions of unity provide a documented route to complex interfaces, including linear Stokes examples[20]. That evidence motivates an extension, but does not validate the present unsteady nonlinear solver on such geometries. A credible extension would preserve boundary and divergence constraints, compare retained dimensions and quadrature work with deterministic local spaces, and repeat both velocity/vorticity qualification and complete-cost timing. Accordingly, our compact-space and kernel conclusions are restricted to the specified separable finite annulus.

Pure radial reparameterization should be distinguished from an angle-dependent geometry. For a smooth invertible map $r = \rho(\eta)$, the metric coefficients remain independent of θ. With the same angularly invariant boundary operators, Fourier orthogonality preserves the implicit modal blocks; only the radial matrices and their conditioning change. The $O(Kd^2)$ independent dense-solve cost remains applicable to the current Galerkin design. A map for an eccentric or noncircular body generally introduces angle-dependent coefficients in this representation and couples angular modes. This observation is about the chosen coordinates and operators, not a theorem that every irregular geometry requires a particular iterative algorithm.

If the coupled implicit matrix A and a fixed preconditioner B are SPD and the assumed spectral-equivalence inequality holds, exact-arithmetic PCG gives the standard energy-error bound[21]

$$\frac{\|e_j\|_A}{\|e_0\|_A} \le 2\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^j, \qquad \kappa = \kappa_2(B^{-1/2} A B^{-1/2}) \le \beta/\alpha. \tag{13.3}$$

For $\kappa > 1$, this supplies a sufficient iteration scale $O(\sqrt{\beta/\alpha}\log(1/\varepsilon))$ for relative energy-error reduction ε; it is not a measured iteration count or a lower bound. The cost also includes applying A, solving with B, and reductions. The bound does not automatically apply to nonsymmetric discretizations, indefinite systems, or variable/inexact preconditioners. Smooth mapping alone has not established the needed α, β for this study. A pseudospectral metric product may use angular FFTs, whereas an assembled sparse coupling or a short Fourier convolution may use other products. Thus neither PCG nor an FFT at every matrix–vector product is mandatory, and the annular timing results cannot predict the new operator balance.

13.2 Synthesis and conclusions

The evidence separates numerical design from implementation. The piecewise schedule removes measured nonlinear evaluations and applies equally to RFM and Legendre. Compact qualifying spaces exist in both basis families. The custom fused operator reduces local nonlinear cost, while the warp reduction removes a substantial observation bottleneck shared by Galerkin and FD. Startup simplification and explicit caching affect different terms of the runtime. None of these facts, individually, establishes an end-to-end advantage from random features.

The closest experimentally supported bottleneck is the mismatch between a short GPU workload and its preparation costs. A historical profile assigns about 0.058 s of a 1.70 s full solve to host spatial preparation, approximately 3.4%. This is a diagnostic for that configuration, not a universal bound, but it provides little motivation for a wholesale language rewrite. Rewriting a measured hot loop can help; replacing high-level orchestration alone does not reduce the number of nonlinear evaluations, library FFTs, or device matrix operations. High-level flow solvers can already combine efficient kernels and numerical libraries [18].

The finite outer boundary is a major limitation. Increasing R from 6 to 8 produces an unshifted maximum vorticity difference of about 50.39% on the common region $r \leq 4$, with accumulated phase differences included. This is a sensitivity result between finite-domain simulations, not a rigorous estimate of exact-solution error. It prevents interpreting the present 0.5% qualification as an infinite-domain cylinder benchmark. The current study also does not validate pressure, drag, lift, three-dimensional turbulence, higher Reynolds numbers, or a hardware-independent speed advantage.

Advanced trained PINN methods were not measured under the same complete field, probe, and timing protocol in this phase. Consequently the paper compares a fixed-feature Galerkin discretization with its deterministic and finite-difference controls, without ranking PINN training methods. Earlier cavity and other flow studies do not automatically inherit the final annulus qualification. Broader claims require new case-specific reference and timing evidence.

The added ablation further shows that deterministic midpoint frequencies with the same analytic layers reach the field tolerance at both resolutions; randomness is not necessary for these configurations. Removing the nonzero-mode layers greatly increases error but also decreases retained rank, preventing an equal-dimensional attribution to a feature family alone. The delayed-switch controls and error profiles clarify accuracy tradeoffs without supplying a new speed ranking.

For the specified $\mathrm{Re} = 100$ finite-annulus problem, all 78 registered GPU qualification trajectories meet the common velocity and vorticity criteria. The two piecewise schedules reduce nonlinear evaluations by 11.11% and 9.36%. Fused FFT pipelines improve local and short-run marching costs but lose on first-use complete cost; the more aggressive nonlinear-update candidates fail accuracy or workload criteria. The final isolated campaign is not accepted, so a stable complete-solve speed advantage over the optimized Legendre and fourth-order spatial FD baselines remains unestablished. The defensible contribution is the explicit mathematical adaptation, kernel implementation, and evidence-based attribution of these distinct effects.

Data and code availability

The original NS evidence corresponds to archived revision `6f2b350`, retaining source, protocols, configuration identities, qualification scores, and failed attempts. Appendix C maps these archived tables and figures to their source records. The repository is publicly accessible; the archived commit and revision inventories identify the corresponding evidence.

Revision-3 evidence is stored separately in `work/rfm_review_v3/`. The `ablation/` directory contains 16 CPU ablation/control trajectories, rescoring of ten frozen GPU controls, 12 capacity-construction checks, and coverage verification. The `profiles_bridge/` directory contains four delayed-switch CPU trajectories, radial and angular error data, cellwise two-reference scores,

and Parseval checks. The `maintenance/` directory contains the explicit-factory maintenance package and seven tests. Protocols, source hashes, outputs, and analysis scripts accompany the revision without overwriting the frozen evidence. These 20 new CPU trajectories do not increase the original GPU qualification count or establish a new isolated performance result. The previous standalone GPU operator example and its 36 correctness cases remain separately identified. The point-by-point response is provided as `response.tex` in the same manuscript project.

Revision-4 diagnostics are under `work/rfm_review_v4/`: the formula audit binds the three shared definitions to their already-correct revision-3 mathematics; the FD audit checks the nonuniform wall closure; and the probe/field analysis tests frequency drift and the limits of schedule-error attribution. Their scripts, input hashes, and verification receipts accompany the release. No additional NS trajectory, reference solution, or timing experiment was produced in revision 4.

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A Notation, scoring grids, and reproduction order

Sections 3 and 4 specify the model, boundary lifting, feature transformations, and weak equations. This appendix collects cross-section notation and the independent reconstruction grids. Input feature count, retained rank, quadrature size, and FFT length describe different dimensions of a solver.

The weights include $r^2\,ds\,d\theta$, and the velocity norm combines u_r, u_θ. The 19 physical checkpoints are 0.5, 1, 2, 4, 6, 8, 8.125, 8.5, 9, 10, 12, 16, 24, 36, 48, 54, 60, 66, and 72. Piecewise schedules additionally save the states at $8+1/\text{late_rate}$ and $8+2/\text{late_rate}$ for implementation consistency. These are not two extra reference-scored physical checkpoints. Probe locations are $(x/D, y/D) = (1.5, 0), (2.5, 0), (3.5, 0)$. Samples are taken at intervals of 0.125, giving 577 times including the zero initial state. Transverse velocity and vorticity at each location give six histories.

Reproduction should first fix the archive identity, source files, and numerical references listed in Appendix C. Check the geometry, seed, retained rank, time rates, and cache policy for each

Table 14: Principal symbols and their computational meaning.

Symbol	Meaning
s, z, L	$s = \log(r/a)$, $L = \log(R/a)$, $z = s/L$
m, d, d_0	Input radial feature count, retained nonzero-mode rank, separate mean rank (30)
K, N, q	Highest retained angular mode, angular FFT length, Galerkin radial quadrature count
n	FD radial intervals; $n+1$ nodes and $n-1$ interior unknowns
M_k, K_k	Nonzero-mode mass and viscous matrices
H, h, ρ	Time steps before and after the bridge and ratio h/H
H_{ref}	Phase whitening matrix $a_{\mathrm{ref}}M + h_{\mathrm{ref}}K$, also denoted H in the analysis
$N_{\mathrm{nl}}, C_{\mathrm{nl}}$	True nonlinear evaluation count and effective cost per evaluation

Table 15: Independent physical-error reconstruction grids.

Grid	Radial points	Angular points	Angular shift
Gauss	123	433	0.23
Uniform midpoint	129	449	0.43
Layer-refined	144 (96/24/24)	457	0.31

configuration. Reconstruct and score each field on every registered grid and time, then evaluate CPU/GPU implementation consistency and the piecewise triangle safeguard. Only qualified candidates proceed to independent timing; concurrent qualification elapsed times are not isolated timings. Batch completeness, terminal records, and external-load checks precede calculation of speed ratios. The revised standalone operator sample is evaluated separately: its checks cover synthetic fixed states and do not replace flow qualification.

B Frozen-vorticity structure and scope

In the real Fourier representation, the mean has weight 1, and the real and imaginary parts of each nonzero mode have weight 2. For frozen vorticity ω^*, the weak pairing is

$$\sum w\, \omega^*(u_\theta v_r - u_r v_\theta).$$

Using the same quadrature and mutually adjoint reconstruction and projection maps, interchanging u and v reverses the sign. Consequently the homogeneous discrete matrix F satisfies $F^{\mathsf{T}} = -F$. This property must be checked at the discrete level.

The update with lifting, forcing, and residual work is

$$\left[\tfrac{3}{2}M + hK - hF\right]x_{n+1} = 2Mx_n - \tfrac{1}{2}Mx_{n-1} + \mathcal{L}_{n+1} + \mathcal{R}_{n+1}. \tag{B.1}$$

Here $\mathcal{L}$ contains boundary lifting and external forcing, and $\mathcal{R}$ is the original left-hand side minus the right-hand side including $\mathcal{L}$. Define

$$\|x\|_M^2 = x^{\mathsf{T}}Mx, \qquad G(x,y) = \|x\|_M^2 + \|2x - y\|_M^2. \tag{B.2}$$

The K-weighted seminorm is defined analogously. At constant step size h, pairing (B.1) with $4x_{n+1}$ gives

$$\begin{aligned} &G(x_{n+1}, x_n) - G(x_n, x_{n-1}) + \|x_{n+1} - 2x_n + x_{n-1}\|_M^2 + 4h\|x_{n+1}\|_K^2 \\ &\quad = 4\langle x_{n+1}, \mathcal{L}_{n+1}\rangle + 4\langle x_{n+1}, \mathcal{R}_{n+1}\rangle. \end{aligned} \tag{B.3}$$

The identity follows by expanding $4\langle x, \frac{3}{2}Mx - 2My + \frac{1}{2}Mz\rangle$ and using skew symmetry to cancel the F contribution. The residual $\mathcal{R}$ is already scaled as in (B.1), so its work term carries no additional factor of h. With homogeneous boundary data, zero forcing, and an exact solve, $\mathcal{L} = \mathcal{R} = 0$ and G is nonincreasing. Driven cases retain the right-hand side. The quantity G describes the homogeneous velocity component. Total kinetic energy also includes the boundary lift, and vorticity and phase errors require separate evaluation.

Let $A = \frac{3}{2}M + hK = LL^{\mathsf{T}}$ and $S = L^{-1}FL^{-\mathsf{T}}$. Then $S^{\mathsf{T}} = -S$. In the symmetrically transformed coordinates, $B = I - hS$ satisfies

$$B^{\mathsf{T}}B = BB^{\mathsf{T}} = I - h^2S^2, \qquad \|Bz\|_2^2 = \|z\|_2^2 + h^2\|Sz\|_2^2.$$

Hence $\sigma_{\min}(B) \geq 1$. This bound concerns the transformed matrix. GMRES iteration count, errors from applying the coordinate transforms, and physical accuracy require separate evaluation. The main text reports the observed coarse-step failures and added FFT workload.

C Archive and verification records

All records below refer to commit `6f2b350`. The labels link to the original files. Path components `v7` and `v8` identify experimental revisions. Theoretical notes, numerical qualification, local timing, and full-batch timing serve distinct purposes.

C1 Case and boundaries `work/rfm_v6/full_case.py`

C2 Spatial spaces and matrices `work/rfm_v6/full_galerkin.py`

C3 Bridge and lifting derivation `work/rfm_v7/theory_sbdf2_single_bridge.txt`

C4 Phase pencils and scope of the anchor result `work/rfm_v8/theory_phase_pencil_scope_note.txt`

C5 Measured phase setup times `work/rfm_v8/galerkin_phase_pencil_audit_canonical_mean.json`

C6 Final 39 configurations and two-cache-policy protocol `work/rfm_v8/rank_fair_cache_qualification_output/protocol.json`

C7 Final 78 numerical qualification runs `work/rfm_v8/rank_fair_cache_qualification_output/qualification_summary.json`

C8 Rank expansion and failed candidates `work/rfm_v8/rank_parity_expansion_output/summary.json`

C9 Fused operator and short-run total cost `work/rfm_v8/fused_march_t9_output/summary.json`

C10 Isolation check for the historical startup improvement `work/rfm_v8/thin_formal_high_verification.json`

C11 Short-run Lamb accuracy and workload `work/rfm_v8/lamb_t16_qualification_output/summary.json`

C12 Disposition of the final full timing campaign `work/rfm_v8/rank_fair_cache_recovery_formal_final_closure/final_closure.json`

Table 5 was recomputed group by group from all 78 `independent_score.json` files. Table 7 reads `metadata.nonlinear_evaluations` from `result.json` in qualification slots `slot_000`, `slot_010`, `slot_026`, and `slot_036`. Table 11 uses fixed-state CUDA Graph event records, and Table 12 uses the local phase-setup audit. Figures 2, 3, and 7 retain the verified plotting and data sources from the same archive. Each performance interpretation applies within the scope of its own timing campaign.

Source hashes are in `supplement/evidence_revision.json`. The previous standalone operator and GPU checks remain in `supplement/operator_sample/`; new CPU experiments and maintenance are identified in Section 9 and the data-availability statement. Directory READMEs give reproduction commands.